

Marginal Contrastive Discrimination for Interventional Density Estimation

Under a Continuous, Multidimensional Treatment

Abstract

We study the estimation of the interventional density of a scalar outcome under a continuous, possibly multidimensional treatment. Under the back-door adjustment, this target reduces to a single conditional density whose conditioning set contains the treatment and every confounder, and which is therefore high-dimensional while the outcome is scalar. Marginal contrastive discrimination (MCD) is tailored to this asymmetry: it replaces the conditional density by a univariate marginal and a binary classifier, so that no density on a high-dimensional space is ever fitted. The resulting plug-in estimator, however, inherits a first-order bias. Removing that bias by the standard one-step correction requires the influence function (IF) of the target, which contains the generalized propensity score, itself a conditional density in the treatment, and precisely the object the contrastive construction was meant to avoid. Our main contribution is to re-express the same correction through the contrastive factorization. The generalized propensity score then disappears from the formula; in exchange, the correction now requires the influence function of the fitted contrast function, whose training sample is built from the data and which therefore behaves unlike the influence function of an ordinarily trained model. We compute this influence function in closed form for a penalized logistic contrast and assemble the IF-corrected estimator.

1 Introduction

A prediction describes what is likely to happen; a causal statement describes what would happen under an intervention. Decisions call for the second. A clinician choosing a dose, a regulator setting an exposure limit, and an analyst evaluating a policy all need the distribution of an outcome under an intervention. Randomized experiments answer such questions by construction, but they are often unavailable, and the causal quantity must then be recovered from observational data, in which the treatment is chosen rather than assigned and the reasons for the choice usually affect the outcome as well.

Most of the causal-inference literature reports a mean: the average treatment effect or, for a treatment ranging over a continuum, the dose-response curve $t \mapsto \mathbb{E}[Y(t)]$, where $Y(t)$ is the potential outcome under treatment level t . A mean, however, collapses a distribution to a single number, and two interventions with equal means may induce very different outcome distributions. Whenever the question concerns the probability of exceeding a threshold, the spread of the response, or any functional of the outcome law beyond its first moment, the relevant object is the full interventional density

$$p_t(y) = f_{Y(t)}(y),$$

the density of $Y(t)$ at the point y . Its semiparametric theory in the discrete-treatment case is due to Kennedy et al. (2023), who derived the efficient influence function and a doubly robust, augmented inverse-probability-of-treatment-weighted (A-IPTW) estimator; representing its nuisance by a conditional normalizing flow and its target by a truncated orthogonal series gives the CNF+TS estimator of the comparison below. Melnychuk et al. (2023) instead represent both the nuisance and the target directly as normalizing flows and A-IPTW-correct the result (INFS+), and report two ablations of their own architecture as further baselines: a covariate-adjusted flow pair without the correction (INFS), and a version with no target flow at all, using the nuisance flow’s conditional density directly as a plug-in, identical, as they note, to the pre-existing conditional normalizing flow baseline (CNF, Trippe and Turner, 2018). Their comparison also includes two uncorrected regression baselines, a shared neural representation with a homoscedastic Gaussian output (TARNET*, Shalit et al., 2017) or a Gaussian-mixture output (MDN, Bishop, 1994), and two nonparametric kernel baselines, a plug-in kernel mean embedding (DKME, Muandet et al., 2021) and an A-IPTW kernel estimator (KDE, Kim et al., 2018). Separately, Ham et al. (2024) impose log-concavity on the counterfactual density, and Martinez-Taboada and Kennedy (2023) avoid an explicit density representation altogether, via kernel Stein discrepancies. All of the above assume a treatment with finitely many levels; Appendix J tabulates every estimator abbreviation used here, and Appendix I every other abbreviation used in the paper. They are not part of the simulation study of Section 7, which is confined to the IF-corrected estimator and its two discriminants (Section 7.2), but a broader comparison against this same list, on the real and semi-synthetic datasets of Melnychuk et al. (2023), is under way in a companion, more applied chapter.

This paper treats the continuous case, which is both more common in applications and harder in theory. Doses, durations, concentrations and continuous policy instruments are not naturally discretized, and applications often involve several such quantities at once, so the treatment is a vector $T \in \mathcal{T} \subseteq \mathbb{R}^d$. With a continuous treatment the propensity score becomes the *generalized* propensity score of Imbens (2000) and Hirano and Imbens (2004), a conditional density in t rather than a probability, and the target is no longer estimable at the parametric rate: Kennedy et al. (2017) showed that the dose-response curve requires smoothing in the treatment argument, and Colangelo and Lee (2020) gave the corresponding debiased machine-learning estimator; more recently, Hudson et al. (2023) study nonparametric inference for the same dose-response curve under weaker overlap. In all of this line of work the target has remained the mean. The density literature runs in the opposite direction: Kennedy et al. (2023) and the baselines listed above target the full counterfactual density, but only for a treatment with finitely many levels. To our knowledge no prior work targets the density under a continuous treatment; here we keep the treatment continuous, and vector-valued, but take the target to be the density.

Our identification rests on the back-door adjustment of Pearl (1995, 2009), which expresses the target as the conditional outcome density averaged over the covariates,

$$p_t(y) = \mathbb{E}_X[f_{Y|T=t,X}(y)].$$

Estimation therefore reduces to a single conditional density, evaluated at the treatment level t and averaged over the covariate distribution. This is where the difficulty concentrates. The conditioning set is the pair (T, X) : it holds the treatment and every confounder, and is high-dimensional, whereas the outcome is scalar. Standard nonparametric conditional-density estimators are poorly suited to that asymmetry, because their rates deteriorate with the dimension of the conditioning argument (Stone, 1980; Tsybakov, 2009), and this rate governs the error of the whole procedure.

Marginal contrastive discrimination (MCD) (Meziani et al., 2025) is designed for exactly this asym-

metry. It writes the conditional density as a univariate marginal times a density ratio, and estimates the ratio by a binary classifier that separates observed vectors from vectors whose outcome has been resampled from its own marginal. No density on a high-dimensional space is ever fitted: the marginal is univariate, and the ratio is recovered by classification, a problem any supervised learner can solve. The same principle underlies noise-contrastive estimation (Gutmann and Hyvärinen, 2010) and classification-based density-ratio estimation (Sugiyama et al., 2012), on which the contrastive construction builds.

Plugging this estimate into the back-door average yields an estimator of $p_t(y)$, but, like every plug-in built on a nonparametric nuisance, it is biased at first order: the bias equals the covariate average of the conditional-density error and vanishes only as slowly as that error (Bickel and Ritov, 1988; Robins et al., 2008). The standard remedy is the one-step correction of semiparametric theory (van der Laan and Rubin, 2006; Chernozhukov et al., 2018; Kennedy, 2022), which adds the empirical average of the target’s influence function (IF) and renders the estimator first-order insensitive to the nuisance. That correction, however, runs into a problem specific to our target. The influence function of $p_t(y)$ contains the generalized propensity score: a second nuisance, and a conditional density in the treatment coordinate, exactly the object the contrastive construction was adopted to avoid. Written in the usual way, the correction gives back the difficulty the estimator was meant to remove.

The resolution exploits a basic fact about influence functions. The influence function of a target is unique, but the target itself admits several equivalent expressions, and differentiating any of them yields an equal, though differently written, formula for the same object. We differentiate the target in its *contrastive* form rather than its back-door form. In that form the generalized propensity score never appears. The price is a new object, the influence function of the fitted contrast function, which behaves unlike the influence function of an ordinarily trained model because the contrastive training sample is itself built from the data.

Contributions. [TO COMPLETE: À compléter.]

Organization. [TO COMPLETE: À compléter.]

2 Setting

This section fixes the estimand and the ingredients used to estimate it. We first record the causal assumptions under which the interventional density is identified from observational data, then form the plug-in estimator from an arbitrary estimator of the conditional density, and show that it is biased at first order. That bias is the object the remainder of the paper is devoted to removing.

2.1 Notation

Let $T \in \mathcal{T} \subseteq \mathbb{R}^d$ be a treatment, $X \in \mathcal{X} \subseteq \mathbb{R}^p$ a vector of covariates and $Y \in \mathcal{Y} \subseteq \mathbb{R}$ a scalar outcome. We observe n independent copies of

$$W = (T, X, Y) \in \mathcal{W} = \mathcal{T} \times \mathcal{X} \times \mathcal{Y}, \quad W \sim F,$$

and assume F has a density with respect to Lebesgue measure. Following Meziari et al. (2025), we write $f_{T,X,Y}$, $f_{T,X}$, f_X , f_Y for the joint and marginal densities and $f_{Y|T=t,X=x}$ for the conditional density of the outcome. For $t \in \mathcal{T}$, let $Y(t)$ be the potential outcome under the intervention setting the treatment to t .

2.2 Causal structure and identification

We assume the causal structure of Figure 1: the covariates X affect both T and Y , the treatment T affects Y , and no unobserved variable is a common cause of T and Y . The only path between T and Y besides the direct one is $T \leftarrow X \rightarrow Y$, and conditioning on X blocks it. This is the back-door criterion of Pearl (2009).

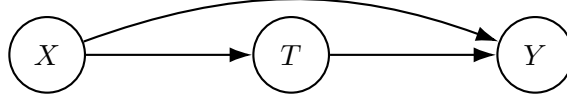


Figure 1: Causal structure assumed throughout. The treatment T is continuous and possibly multidimensional, the outcome Y is scalar, and the covariates X affect both. The only back-door path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X .

To turn this structure into identification of the potential outcomes $\{Y(t) : t \in \mathcal{T}\}$, we impose the standard conditions for continuous treatments, due to Imbens (2000) and Hirano and Imbens (2004).

Assumption 1 (Identification). Let $t \in \mathcal{T}$ be the treatment level of interest.

- (i) *Consistency and no interference.* $Y = Y(T)$, and a unit's potential outcomes do not depend on the treatment levels of other units.
- (ii) *Weak unconfoundedness.* $Y(t) \perp T \mid X$.
- (iii) *Positivity.* There exists $e_{\min} > 0$ such that $f_{T|X=x}(t) \geq e_{\min}$ for every x with $f_X(x) > 0$.

The three parts play distinct roles. Consistency ties the potential outcomes to the observed data, equating Y with the potential outcome at the treatment actually received. Weak unconfoundedness is the distributional counterpart of the absent common cause of T and Y in Figure 1: given the covariates, the treatment carries no further information about the potential outcome. Positivity requires the level t to be reachable at every covariate value; its lower bound $e_{\min}$ is what will keep the correction of Section 3 finite. Under these conditions $Y(t)$ is identified, because weak unconfoundedness makes the conditional density of Y at $T = t$ equal to the density of $Y(t)$ at each covariate value, and averaging over the covariates recovers the interventional law.

Definition 1 (Interventional density). For $t \in \mathcal{T}$ and $y \in \mathcal{Y}$, the interventional density is

$$p_t(y) := f_{Y(t)}(y) = \mathbb{E}_X[\eta(X)] = \int_{\mathcal{X}} \eta(x) f_X(x) dx, \quad \eta(x) := f_{Y|T=t, X=x}(y), \quad (1)$$

where $\eta(x)$ is the conditional outcome density at the point (t, x, y) ; we suppress its dependence on the fixed t and y .

Two features of this estimand matter later. First, the covariate is averaged out while t and y are fixed evaluation points; this asymmetry is what produces the point masses in the influence function. Second, the only unknown is the conditional density η , since the covariate distribution is handled by the empirical distribution of $X_1, \dots, X_n$. Its conditioning set (T, X) has dimension $d + p$ while the outcome is scalar, and it is this imbalance that makes standard estimators inaccurate and motivates the estimator used below.

The correction of Section 3 will also involve the conditional law of the treatment, which we introduce here.

Definition 2 (Generalized propensity score). The conditional density of the treatment given the covariates,

$$e(t \mid x) := f_{T|X=x}(t), \quad (2)$$

is the generalized propensity score of Imbens (2000) and Hirano and Imbens (2004).

For a continuous treatment $e(t \mid x)$ is a density in $t \in \mathbb{R}^d$, not a probability, and part (iii) of Assumption 1 bounds it below. It plays no role in identification, the estimand (1) does not mention it, and its cost as a nuisance is precisely what the contrastive route of Section 4 will save.

2.3 The plug-in estimator and its bias

Let $\hat{\eta}$ be any estimator of the conditional density η , the marginal contrastive discrimination estimator of Section 4 is one choice, but nothing in this subsection uses it. Averaging $\hat{\eta}$ over the covariates, as Definition 1 prescribes, gives the plug-in estimator

$$\hat{p}_t^{\text{pi}}(y) = \frac{1}{n} \sum_{i=1}^n \hat{\eta}(X_i). \quad (3)$$

To see why this estimator is biased, suppose $\hat{\eta}$ is fitted on a sample independent of $X_1, \dots, X_n$, and split the error by adding and subtracting the covariate average of the true η :

$$\hat{p}_t^{\text{pi}}(y) - p_t(y) = \underbrace{\frac{1}{n} \sum_{i=1}^n (\eta(X_i) - p_t(y))}_{\text{sampling error}} + \underbrace{\frac{1}{n} \sum_{i=1}^n (\hat{\eta}(X_i) - \eta(X_i))}_{\text{nuisance error}}. \quad (4)$$

The two pieces behave in opposite ways. The sampling error averages the fixed function $x \mapsto \eta(x) - p_t(y)$, which has mean zero because $\mathbb{E}_X[\eta(X)] = p_t(y)$; it is therefore centred and of order $n^{-1/2}$, the ordinary fluctuation of a sample mean.

The nuisance error behaves differently. Because $\hat{\eta}$ is fitted on a separate sample, we may treat it as fixed and take expectations over $X_1, \dots, X_n$ alone. Each term then has the same expectation, so

$$\mathbb{E} \left[\hat{p}_t^{\text{pi}}(y) - p_t(y) \mid \hat{\eta} \right] = \mathbb{E}_X [\hat{\eta}(X) - \eta(X)], \quad (5)$$

the average error of the conditional-density estimate. Nothing forces this quantity to vanish: if $\hat{\eta}$ over- or under-estimates η on average, so does the plug-in. The offset is a property of the back-door average itself, shared by every conditional-density estimator, not an accident of any particular $\hat{\eta}$.

The right-hand side of (5) is what the semiparametric literature calls the first-order bias of the plug-in, and the name deserves a word of caution, since it is not the bias of an estimator in the elementary sense. The finite-sample bias $\mathbb{E}[\hat{\theta}] - \theta$ at a fixed model would vanish as the nuisance became consistent; the quantity in (5) is a different object, namely the leading term of the estimation error expanded in the nuisance error $\hat{\eta} - \eta$, and it is of first order precisely because it is linear in that error. The same term recurs across the literature under names that stress different facets of it: an obstruction to efficient estimation with an infinite-dimensional nuisance (Bickel and Ritov, 1988; Robins et al., 2008), a regularization bias inherited from the smoothing that any flexible estimator applies (Chernozhukov et al., 2018), or the plug-in bias that the one-step correction is built to remove (van der Laan and Rubin, 2006; Kennedy, 2022). We adopt the term first-order bias for the object (5).

A larger sample does not cure this bias. The nuisance error shrinks with n , but only at the rate of $\|\hat{\eta} - \eta\|$, and for a conditional density on a conditioning set of dimension $d + p$ that rate is slower than $n^{-1/2}$ for any flexible estimator: nonparametric conditional-density estimators have rates that deteriorate with the dimension of the conditioning argument (Stone, 1980; Tsybakov, 2009; Bashtannyk and Hyndman, 2001; Hall et al., 2004), and with several treatments and confounders this error dominates everything else. On the $\sqrt{n}$ scale the first-order bias therefore diverges rather than settling, so $\sqrt{n}(\hat{p}_t^{\text{pi}}(y) - p_t(y))$ has no centred limit and yields no valid confidence interval. The obstruction is not a small sample but the sub-parametric rate of conditional-density estimation in high dimension, which no sample size and no choice of estimator overcomes.

For the interventional density the consequence is more damaging than for a mean. The density is estimated pointwise in y , so the first-order bias is not one mislocated number but a systematic distortion of the whole curve $y \mapsto \hat{p}_t^{\text{pi}}(y)$. The very quantities that motivate estimating a density, tail probabilities, dispersion, the chance of exceeding a threshold, are functionals of this curve, and each inherits the distortion. Correcting the first-order bias is thus a prerequisite for the inferences the interventional density is meant to support, not a cosmetic refinement.

The remedy is to make the nuisance enter only at second order. A second-order term is quadratic in $\hat{\eta} - \eta$, of order $\|\hat{\eta} - \eta\|^2$, which is $o(n^{-1/2})$ as soon as the nuisance converges faster than $n^{-1/4}$, a requirement far weaker than the parametric rate. The device that achieves this is the one-step correction, to which we now turn.

3 First-order correction

The plug-in of Section 2.3 carries the first-order bias (5), and the previous section argued that removing it is necessary before any inference. The one-step correction does so by adding to the plug-in the empirical average of an influence function, chosen so that the first-order term cancels and the nuisance enters only at second order. To build it we need the influence function of $p_t(y)$. We first recall the von Mises expansion that defines it, then compute it, and then confront the difficulty it raises: the correction it produces is valid, but it reintroduces the generalized propensity score. That difficulty is what motivates Section 4.

3.1 Von Mises expansion

View the target as a functional $\psi(F) = p_t(y)$ of the data law. To probe its sensitivity to the data, perturb F towards a point mass at $w \in \mathcal{W}$,

$$F_\varepsilon = (1 - \varepsilon)F + \varepsilon \delta_w, \quad \varepsilon \in [0, 1], \quad (6)$$

and differentiate the target along this path. The derivative is the influence function, in the sense of Hampel (1974) and the semiparametric development of Bickel et al. (1993).

Definition 3 (Influence function). The influence function of ψ at F in the direction δ_w is

$$\text{IF}(w; \psi, F) := \left. \frac{\partial}{\partial \varepsilon} \psi(F_\varepsilon) \right|_{\varepsilon=0}.$$

Because $\varepsilon \mapsto \psi(F_\varepsilon)$ is an ordinary real-valued map, its derivative obeys the ordinary sum, product, quotient and chain rules. We collect these rules, in the form adapted to IF, in Appendix A; they are all the machinery the computations below require. The influence function has mean zero,

$\mathbb{E}_F[\text{IF}(W; \psi, F)] = 0$, and it governs the first-order response of the target to a change of law: for any competing law F' ,

$$\psi(F') - \psi(F) = \mathbb{E}_{F'}[\text{IF}(W; \psi, F)] + \text{Rem}_2(F', F), \quad (7)$$

with a remainder of second order in $F' - F$. This expansion is what makes bias removal possible. Reading F' as the true law and F as the estimated one, the plug-in error is the first-order term plus a second-order remainder; adding the empirical average of the estimated influence function to the plug-in cancels that first-order term and leaves only the remainder (Bickel et al., 1993; van der Vaart, 1998; van der Laan and Rubin, 2006; Chernozhukov et al., 2018). It remains to compute $\text{IF}(w; p_t(y), F)$, which is the task of the next subsection.

3.2 The influence function of the interventional density

We apply Definition 3 to the target. Since $p_t(y)$ is an integral over x of a ratio of densities in (t, x, y) , differentiating it along the path (6) with the calculus of Appendix A yields the following; the full derivation is in Appendix B.2.

Proposition 1 (Nonparametric correction). *Under Assumption 1 and the regularity conditions of Appendix B.1, the influence function of $p_t(y)$ at F , evaluated at $w = (t_w, x_w, y_w)$, is*

$$\text{IF}(w; p_t(y), F) = \frac{\delta_{t_w}(t)}{e(t \mid x_w)} \left(\delta_{y_w}(y) - \eta(x_w) \right) + \eta(x_w) - p_t(y), \quad (8)$$

where δ_a denotes the Dirac measure at a and $\eta(x_w) = f_{Y|T=t, X=x_w}(y)$.

The shape of (8) is exactly the shape anticipated after Definition 1, and reading it that way explains each term rather than merely labelling it. Because the covariate is averaged out, a contamination at x_w enters the target smoothly, and its effect is the ordinary centred comparison $\eta(x_w) - p_t(y)$ between the conditional density at the contaminating covariate and its average, the same term one would obtain for any smooth average. Because the treatment and outcome are read off at fixed points, by contrast, a contamination that lands on those points has a singular effect: it can only register through Dirac masses in t and y , which is the first term. The weight $1/e(t \mid x_w)$ on that term is the inverse generalized propensity score, and it is there for a familiar reason, a contamination at treatment level t_w is informative about the level t only in proportion to how rarely that level occurs at x_w , so its influence is inflated when $e(t \mid x_w)$ is small. The essential observation for what follows is simply that e appears at all: an object absent from the target (1) has entered its influence function.

Two limiting cases place (8) in the literature and confirm it. For a discrete treatment the Dirac in t becomes an indicator, the generalized propensity score becomes an ordinary propensity score, and the expression reduces to the influence function of Kennedy et al. (2023). For the mean $\mathbb{E}[Y(t)]$ the outcome is integrated rather than evaluated, the Dirac in y disappears, and the expression reduces to that of Kennedy et al. (2017). Our target combines a continuous treatment with a pointwise outcome, and both singular features appear together; to our knowledge this influence function has not been derived before in that combination.

The Dirac masses are not estimable, so we regularize them by kernel smoothing with bandwidths h_T and h_Y ; the substitution and its bias are carried out in Appendix B.3. Feeding the smoothed influence function into the one-step construction of (7), the plug-in (3) plus the empirical average of the smoothed influence function, gives the *classical estimator* below; the plug-in term cancels

algebraically against a matching term inside that average (Appendix B.3, display (44)) and so does not appear in it explicitly.

$$\hat{p}_t^c(y) = \frac{1}{n} \sum_{i=1}^n \left[\frac{K_{h_T}(T_i - t)}{\hat{e}(t | X_i)} \left(K_{h_Y}(Y_i - y) - \hat{\eta}(X_i) \right) + \hat{\eta}(X_i) \right]. \quad (9)$$

Smoothing enters only through the two point evaluations; the nuisances e and η are left unsmoothed and may be fitted by any method. This estimator removes the first-order bias of the plug-in, and does so robustly, in the following precise sense; the remainder is bounded in Appendix B.4.

Proposition 2 (Double robustness). *Under the conditions of Appendix B.4, the second-order remainder of the estimator (9) factorizes as a product of the errors of e and η . Consequently $\hat{p}_t^c(y)$ is $\sqrt{n}$ -consistent whenever each nuisance converges at rate $n^{-1/4}$, and one nuisance may converge more slowly if the other converges faster.*

This proposition deserves unpacking, because the mechanism behind it is precisely what the contrastive route of Section 4 gives up, and half of Section 8.1 is an accounting of that loss. The name the discrete-treatment literature gives (9), *augmented* inverse-probability-of-treatment weighting, describes its construction literally: an inverse-probability weighted estimator to which one adds an augmentation term built from an outcome model. Three estimators of the same target $p_t(y) = \mathbb{E}_X[\eta(X)]$ sit side by side, differing only in which nuisances they use.

The plug-in (3) averages a fitted outcome model over the covariates and uses no treatment model at all. The inverse-probability-weighted (IPW) estimator does the opposite, and it repays deriving rather than quoting, because the change of measure behind it is the same one that reappears, hidden, inside the augmentation below. Under Assumption 1, weighting the observed law by the inverse generalized propensity score undoes the dependence of the treatment on the covariates:

$$\begin{aligned} \mathbb{E} \left[\frac{\delta_T(t)}{e(t | X)} \delta_Y(y) \right] &= \iiint f_{T,X,Y}(\tau, x, u) \frac{\delta_\tau(t)}{e(t | x)} \delta_u(y) \, d\tau \, dx \, du \\ &= \int \frac{f_{T,X}(t, x)}{e(t | x)} f_{Y|T=t, X=x}(y) \, dx = \int f_X(x) \eta(x) \, dx = p_t(y), \end{aligned} \quad (10)$$

the last step using the factorization $f_{T,X}(t, x) = f_X(x) e(t | x)$: the weight cancels the density of the assignment mechanism exactly, and what survives is the covariate average that defines the target. The two Dirac masses in (10) are the same ones that appear in the influence function (8), and they are regularized the same way, by K_{h_T} and K_{h_Y} ; replacing the expectation by a sample average and e by an estimate then gives

$$\hat{p}_t^{\text{ipw}}(y) = \frac{1}{n} \sum_{i=1}^n \omega_i K_{h_Y}(Y_i - y), \quad \omega_i := \frac{K_{h_T}(T_i - t)}{\hat{e}(t | X_i)}. \quad (11)$$

It uses no outcome model at all: it reweights the observed sample so that the treatment behaves as though it had been randomized. Each unit contributes its own outcome, smoothed by K_{h_Y} and weighted by how close its treatment is to t divided by how likely it was to receive that treatment given its covariates, so that units unlikely to be treated at t are correspondingly up-weighted. That last feature is also its weakness: where $e(t | x)$ is small the weight is large, a handful of observations dominate the average, and the variance degrades — the more so here, since K_{h_T} concentrates the sum on the few units whose treatment lands within h_T of t , and a vector treatment makes those

few fewer still. The classical estimator (9) is the third, and regrouping its summand makes the construction visible:

$$\hat{p}_t^c(y) = \underbrace{\frac{1}{n} \sum_{i=1}^n \hat{\eta}(X_i)}_{\text{plug-in}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \omega_i (K_{h_Y}(Y_i - y) - \hat{\eta}(X_i))}_{\text{augmentation}}, \quad (12)$$

the plug-in, plus inverse-probability weighting applied to the *residuals* of the outcome model rather than to the outcomes themselves. Table 1 collects the three side by side.

Estimator	Formula	Nuisances	Consistent if
Plug-in (back-door)	$\frac{1}{n} \sum_i \hat{\eta}(X_i)$	$\hat{\eta}$ alone	$\hat{\eta}$ correct
IPW	$\frac{1}{n} \sum_i \omega_i K_{h_Y}(Y_i - y)$	$\hat{e}$ alone	$\hat{e}$ correct
A-IPTW	$\frac{1}{n} \sum_i [\omega_i (K_{h_Y}(Y_i - y) - \hat{\eta}(X_i)) + \hat{\eta}(X_i)]$	$\hat{e}$ and $\hat{\eta}$	$\hat{e}$ or $\hat{\eta}$ correct

Table 1: Three estimators of $p_t(y) = \mathbb{E}_X[\eta(X)]$, with $\omega_i = K_{h_T}(T_i - t)/\hat{e}(t | X_i)$. The first two each rest on a single nuisance and fail when it fails; the third uses both and tolerates the failure of either one. The last column states consistency up to the smoothing bias of K_{h_T} and K_{h_Y} , which is a separate error, bounded in Appendix B.3.

Why the augmentation buys robustness can be read off (12) directly: $\hat{\eta}$ enters twice with opposite signs, while $\hat{e}$ enters only as a multiplier of a residual. Take each nuisance in turn, in the population and with the smoothing bias set aside.

Suppose first that $\hat{\eta}$ is correct. Because the weight carries $K_{h_T}(T_i - t)$, the residual is evaluated where T is close to t , and there $\mathbb{E}[K_{h_Y}(Y - y) | T = t, X] = \eta(X)$: the augmentation averages a residual of mean zero and vanishes in expectation. The estimator collapses onto the plug-in, which is consistent. A wrong $\hat{e}$ does no damage, because it multiplies a quantity that is already centred.

Suppose instead that $\hat{e}$ is correct. Then $\mathbb{E}[\omega | X] = 1$, so the weights integrate to one within each covariate stratum. The first half of the augmentation recovers the target, $\mathbb{E}[\omega K_{h_Y}(Y - y)] = p_t(y)$, and the second satisfies $\mathbb{E}[\omega \hat{\eta}(X)] = \mathbb{E}_X[\hat{\eta}(X)]$, which is exactly the plug-in term with the opposite sign. The estimator returns $p_t(y) + \mathbb{E}_X[\hat{\eta}] - \mathbb{E}_X[\hat{\eta}] = p_t(y)$: a wrong $\hat{\eta}$ cancels between its two occurrences.

Either nuisance may therefore be wrong, provided the other is not, which is Proposition 2 in qualitative form; the quantitative form, the product structure of the remainder, is what additionally permits one to converge slowly when the other converges fast. When both are correct the augmentation does one thing more: subtracting $\hat{\eta}(X_i)$ before weighting acts as a control variate for the weighted average that (11) leaves bare, reducing its variance, and the estimator attains the semiparametric efficiency bound. This is the structure that Section 8.1 reports as lost once the correction is recomputed through the contrastive factorization: there, no pair of nuisances remains to play these two roles against each other.

3.3 The cost of the correction

The estimator (9) works, but it comes at a price, and naming the price is what motivates the rest of the paper. The correction has introduced a second nuisance, the generalized propensity score e . For a vector treatment e is a conditional density on $\mathbb{R}^d$, exactly the kind of object whose rate degrades with the dimension of the conditioning set, as Section 2.3 discussed for η itself. The correction has

therefore reintroduced, as a second nuisance, the very difficulty that a well-chosen estimator of η would need to sidestep in the first place. Double robustness would let one nuisance be slow if the other were fast, but here e and η are both conditional densities in the treatment coordinate, and both are slow, so the guarantee is empty in the regime that matters.

There is a way out, and it has two steps. The first is to choose the estimator of η so that no density on $\mathcal{T} \times \mathcal{X}$ is ever fitted, which is the subject of the next subsection. The second is to notice that such an estimator does not merely estimate η but rewrites it, and that the target can then be differentiated through that rewriting rather than through the back-door form (1); that is Section 4. The influence function is unique, but the form one differentiates is not, and the second form does not contain e .

3.4 Marginal contrastive discrimination

The diagnosis above is a specification for the estimator of η we should want: one that never fits a density on $\mathcal{T} \times \mathcal{X}$ in the first place, so that the correction has no high-dimensional conditional density to reintroduce. Marginal contrastive discrimination (Meziani et al., 2025) is such an estimator, and it is the one we adopt for the rest of the paper. This subsection introduces it on its own terms, as a conditional-density estimator; Section 4 then puts its particular construction to a use it was not designed for.

Its central idea is to turn the estimation of η , a conditional density on a $(d + p)$ -dimensional conditioning set, into two easier tasks: a univariate density estimation problem, for the marginal f_Y , and a binary classification problem, whose solution recovers the density ratio between η and f_Y . On problems of this shape the method is competitive with state-of-the-art conditional-density estimators at lower cost; we refer to Meziani et al. (2025) for those comparisons. We use it with conditioning variable (T, X) and target variable Y . It starts from the elementary factorization

$$f_{Y|T=t, X=x}(y) = f_Y(y) \cdot R(t, x, y), \quad R(t, x, y) := \frac{f_{T, X, Y}(t, x, y)}{f_{T, X}(t, x) f_Y(y)}, \quad (13)$$

valid when $f_{T, X}(t, x) > 0$ and $f_Y(y) > 0$. The first factor is a univariate marginal; the second is a density ratio that equals one when Y is independent of (T, X) and otherwise measures their dependence. The classifier learns this ratio.

Definition 4 (Marginal contrast function, $\text{MCFunct}(\pi)$). For a mixture ratio $\pi \in (0, 1)$,

$$D_\pi(t, x, y) = \frac{\pi f_{T, X, Y}(t, x, y)}{\pi f_{T, X, Y}(t, x, y) + (1 - \pi) f_{T, X}(t, x) f_Y(y)}. \quad (14)$$

Definition 4 is best read as a posterior probability. Consider the experiment in which a label $Z \in \{0, 1\}$ is drawn with $\mathbb{P}(Z = 1) = \pi$; when $Z = 1$ one records an observed vector $W = (T, X, Y)$, and when $Z = 0$ one records $W = (T, X, \tilde{Y})$ with $\tilde{Y} \sim f_Y$ drawn independently of (T, X) . Then $D_\pi(t, x, y)$ is exactly $\mathbb{P}[Z = 1 \mid W = (t, x, y)]$, so recovering the contrast function is a supervised classification problem: tell a genuine triple from one whose outcome has been swapped for an independent draw. In practice one builds a contrastive dataset $\mathcal{D}_N^{W, Z} = \{(W_i, Z_i)\}_{i=1}^N$, the condition $\text{MDCond}(\pi)$ of Meziani et al. (2025) — by labelling each observed vector $Z = 1$ and adjoining vectors with resampled outcomes labelled $Z = 0$.

The classifier and the density ratio encode the same information, so either recovers the other, and hence the conditional density. This is the content of the next identity, which we use repeatedly.

Proposition 3 (Reconstruction identity). *For all (t, x, y) with $f_{T,X}(t, x) > 0$ and $f_Y(y) > 0$,*

$$f_{Y|T=t, X=x}(y) = f_Y(y) R(t, x, y), \quad R(t, x, y) = \frac{1 - \pi}{\pi} \frac{D_\pi(t, x, y)}{1 - D_\pi(t, x, y)}. \quad (15)$$

This is Lemma 1 of Meziani et al. (2025), specialized to conditioning variable (T, X) ; we reproduce the short proof in Appendix C.1 for completeness. The mixture ratio π is a design parameter: different values give different contrast functions but the same density ratio, provided the value used to build $\mathcal{D}_N^{W,Z}$ is the one substituted into (15).

A contrast estimate $\hat{D}_{\text{MCD}}$ and a marginal estimate $\hat{f}_Y$ then combine, through the second identity in (15), into an estimate of the conditional density,

$$\hat{\eta}(x) = \hat{f}_Y(y) \hat{R}(t, x, y), \quad \hat{R} = \frac{1 - \pi}{\pi} \frac{\hat{D}_{\text{MCD}}}{1 - \hat{D}_{\text{MCD}}}. \quad (16)$$

Nothing forces $\hat{\eta}$ to integrate to one in y ; Meziani et al. (2025) restore this by a rescaling step, which we adopt. Substituting (16) into the plug-in (3) already gives a usable estimator of $p_t(y)$, and it is the one the first-order bias of Section 2.3 applies to; everything that follows is about correcting it.

Two features of the construction are worth flagging now, because the whole of Section 4 turns on them, one favourably and one not. Favourably: because the classifier only ever compares observed vectors with recombined ones, it works on the support of the joint law rather than the ambient space, which is why the construction stays feasible when $d + p$ is large (Meziani et al., 2025) — and why η is recovered without any density on $\mathcal{T} \times \mathcal{X}$ ever being fitted. Unfavourably: the negative class is built from the data by recombination, so the classifier’s training law is itself a functional of F . Perturbing F therefore perturbs what the classifier was trained on, and its influence function will not be the textbook one.

4 Correcting the plug-in through the MCD factorization

Section 3 produced a correct correction and then showed its cost, and Section 3.4 committed to an estimator of η chosen so that the cost need not be paid. The two halves now meet. The observation this section rests on is that marginal contrastive discrimination does not merely *estimate* η ; it *rewrites* it, as (13), into a univariate marginal times a density ratio. That rewriting is a new expression for the target, and the target can be differentiated through it.

The influence function is unique, but the form one differentiates is not. Section 3 differentiated the target in its back-door form (1), in which the conditional density appears as a single object, and the derivative produced the generalized propensity score. Substituting (13) instead gives the same functional written in terms of f_Y and R , and differentiating *that* gives the same influence function in a different, equal expression — one in which, as Section 4.2 shows, e does not appear at all. This is the paper’s central move, and it is available only because the nuisance estimator has this particular structure. The rest of the section builds the estimator that the new expression supports, and computes the one object it requires.

That object is the price. It is worth setting out the balance sheet before the derivations, since both sides of it recur throughout the paper.

- *What is gained.* The generalized propensity score disappears from the correction, and with it the d -dimensional kernel K_{h_T} that the classical estimator (9) needs in order to use it: the

corrected estimator smooths only in the outcome coordinate. No density on $\mathcal{T} \times \mathcal{X}$ is fitted at any point, by the plug-in or by the correction. Section 8.2 argues that this is a difference in kind rather than in constant, and that it grows with d .

- *What is paid.* Three things. The correction now requires the influence function of the *fitted contrast function*, and because the contrastive training sample is built from the data, this is not the classical sandwich formula but a three-channel object (Section 4.4) that must be derived from scratch. Double robustness is lost: the two legs of the identified functional have been replaced by a single fitted object, so no second nuisance remains whose good behaviour could rescue a poor contrast fit (Section 8.1). And the correction is confined to the span of the contrast model’s own Jacobian, so it cannot repair a misspecification lying outside it.

The order of what follows is: the data objects the construction needs (Section 4.1), the contrastive form of the influence function and the estimator it supports (Sections 4.2 and 4.3), why the contrast function’s own influence function is not standard (Section 4.4), its derivation for an arbitrary differentiable discriminant, in closed form, together with the resulting algorithm (Sections 4.5–4.8). Sections 5 and 6 are then applications of the general derivation to two specific discriminants.

4.1 Data and cross-fitting

We first fix the data objects the construction uses. Let

$$\mathcal{D}_{\text{obs}}^{T,X,Y} := \{W_i = (T_i, X_i, Y_i) : i = 1, \dots, n\}$$

denote the original observations. In the cross-fitted procedure, we write

$$\mathcal{D}_{\text{obs}}^{T,X,Y} = \mathcal{D}_{\text{fit}}^{T,X,Y} \cup \mathcal{D}_{\text{eval}}^{T,X,Y},$$

where $\mathcal{D}_{\text{fit}}^{T,X,Y}$ is used to estimate the nuisance functions and $\mathcal{D}_{\text{eval}}^{T,X,Y}$ is reserved for evaluating their influence functions; the roles are exchanged and the two resulting estimates averaged. Applying the contrastive construction of Section 3.4 to $\mathcal{D}_{\text{fit}}^{T,X,Y}$ gives

$$\mathcal{D}_{\text{ctr}}^{W,Z} := \{(W_k, Z_k) : k = 1, \dots, N_{\text{ctr}}\}, \quad Z_k \in \{0, 1\},$$

where $Z_k = 1$ denotes a genuine observation and $Z_k = 0$ a recombined one. Both discriminants of Sections 5 and 6 use this single contrastive training set. The marginal density is estimated from the original outcomes in $\mathcal{D}_{\text{fit}}^{T,X,Y}$,

$$\hat{p}_Y \leftarrow \left\{ Y_i : W_i \in \mathcal{D}_{\text{fit}}^{T,X,Y} \right\},$$

the recombined rows in $\mathcal{D}_{\text{ctr}}^{W,Z}$ being used for classification, not as additional observations in this estimate. In the original MCD implementation, where no outer cross-fitting is used, the corresponding kernel density estimate uses all original outcomes $\{Y_i\}_{i=1}^n$.

4.2 The contrastive form of the correction

We begin with the central result of the paper: the same influence function, recomputed from the contrastive factorization, no longer contains the generalized propensity score. Substituting the reconstruction identity (15) into the target (1) factorizes it into a univariate marginal and a covariate-averaged density ratio,

$$p_t(y) = f_Y(y) m(t, y), \quad m(t, y) := \mathbb{E}_X [\mathbf{R}(t, X, y)]. \quad (17)$$

Both factors are functionals of F , and both are estimated from the same data, so a perturbation of F moves them together. Differentiating this product with the calculus of Appendix A gives the following; the derivation is in Appendix C.2.

Proposition 4 (Contrastive form). *Under the conditions of Appendix C, with D_π evaluated at (t, X, y) inside the expectations,*

$$\begin{aligned} \text{IF}(w; p_t(y), F) = f_Y(y) & \left\{ R(t, x_w, y) - m(t, y) + \frac{1 - \pi}{\pi} \mathbb{E}_X \left[\frac{\text{IF}(w; D_\pi, F)}{(1 - D_\pi)^2} \right] \right\} \\ & + m(t, y) (\delta_{y_w}(y) - f_Y(y)). \end{aligned} \quad (18)$$

Each group of terms in (18) has a clear provenance, and knowing where it comes from is what makes the formula usable. The first two terms are the ordinary centred contribution of the covariate average, now written for the density ratio rather than the conditional density directly. The bracketed expectation carries the influence function of the contrast function itself, and appears because the density ratio is not observed but fitted; it is the whole subject of Section 4.4. The final term is the correction attached to the univariate marginal, and it is the only place where a point evaluation survives: the outcome is read off at y inside $f_Y(y)$, and the corresponding Dirac is regularized by K_{h_Y} exactly as in Section 3.2. Crucially, no Dirac appears in the treatment argument, because the contrast function is evaluated at (t, X, y) through a fitted model rather than through a point mass of the data.

The generalized propensity score is absent from (18). The conditional outcome density is still present, by the reconstruction identity of Proposition 3, $f_Y(y) R(t, x_w, y)$ is exactly that density, but it enters only through the two factors of (13), a univariate marginal and a contrast function, neither of which is a conditional density in the treatment coordinate. The influence function is unique, so this contrastive form and the back-door form (8) are two expressions of the same object; we work from the contrastive one, in which the propensity score never appears, and build the IF-corrected estimator around it.

4.3 The IF-corrected estimator

The contrastive influence function of Proposition 4 is all that the one-step construction of Section 3.1 needs. Adding the empirical average of the estimated influence function to the plug-in, and regularizing the one surviving Dirac by K_{h_Y} as in Section 3.2, gives the *IF-corrected estimator*

$$\widehat{p}_t^{\text{ifc}}(y) = \frac{1}{n} \sum_{i=1}^n \left\{ \widehat{f}_Y(y) \left[\widehat{R}(t, X_i, y) - \widehat{m}(t, y) + \frac{1 - \pi}{\pi} \frac{1}{n} \sum_{j=1}^n \frac{\widehat{\text{IF}}(W_i; D_\pi(t, X_j, y))}{(1 - \widehat{D}_{\text{MCD}}(t, X_j, y))^2} \right] + \widehat{m}(t, y) K_{h_Y}(Y_i - y) \right\}, \quad (19)$$

with $\widehat{m}(t, y) = n^{-1} \sum_j \widehat{R}(t, X_j, y)$. Two properties are already visible, and together they are the payoff of the whole contrastive route: the estimator uses no generalized propensity score, and its only bandwidth is h_Y , tied to the point evaluation of the outcome, so no smoothing is performed in the treatment coordinate.

Only one ingredient remains unspecified: $\widehat{\text{IF}}(W_i; D_\pi)$, the influence function of the fitted contrast function. Everything else in (19) is supplied by the MCD density estimator of Section 3.4, the marginal $\widehat{f}_Y$, the density ratio $\widehat{R}$, and their covariate average $\widehat{m}$. That last object is not a standard model influence function, because the classifier is trained on a sample built from the data, and

estimating it is the subject of the rest of this section. Writing down (19) in full therefore requires computing $\text{IF}(w; D_\pi, F)$ for the classifier at hand, and its expression depends on that classifier: Sections 5 and 6 carry out this computation in turn for penalized logistic regression and for a fixed-architecture multilayer perceptron.

4.4 The influence function of the contrast estimate

The IF-corrected estimator (19) hinges on the one object we deferred, $\text{IF}(w; D_\pi, F)$, the derivative of the fitted contrast function with respect to the data-generating law. Influence functions of trained models are not new, and it helps to say why the standard formula does not apply here. For penalized M -estimators the influence function is the classical sandwich expression, treated in modern form by Avella-Medina (2017). In the machine-learning literature Koh and Liang (2017) reintroduced influence functions as a tool for interpreting trained predictors, and their behaviour in that setting was studied by Basu et al. (2021) and Bae et al. (2022). Every one of these formulas describes the effect of perturbing a *fixed* training distribution.

The contrastive construction breaks that assumption. The negative class of $\mathcal{D}_N^{W,Z}$ is generated from the observed data, so the law of the training sample depends on F through two of its marginals. Perturbing F therefore perturbs the training distribution as well as the evaluation distribution, and the derivative acquires terms the classical formula does not have. To make this precise, write the two component laws of the contrastive experiment as

$$P_1(F) := F_{T,X,Y}, \quad P_0(F) := F_{T,X} \otimes F_Y, \quad (20)$$

and let $\mathbb{E}_{P_\pi} := \pi \mathbb{E}_{P_1} + (1 - \pi) \mathbb{E}_{P_0}$ be expectation under the mixture prescribed by $\text{MDCond}(\pi)$. The positive law $P_1(F)$ is F itself, but the negative law $P_0(F)$ is a *product* of two marginals of F , and a contamination of F perturbs each factor of that product separately. It is this product structure that produces the extra channels, as each discriminant treated below will show explicitly.

4.5 A common differentiable classifier form

Everything that follows is derived once, here, for an arbitrary differentiable discriminant. Sections 5 and 6 are then applications of this one derivation, and contain only what is specific to each. Let $F_\beta : \mathcal{W} \rightarrow \mathbb{R}$ be a differentiable logit and write

$$D_\beta(w) = \sigma(F_\beta(w)), \quad J_\beta(w) := \nabla_\beta F_\beta(w), \quad (21)$$

with $\sigma(u) = (1 + e^{-u})^{-1}$. The penalized contrastive risk, its gradient and its minimizer are

$$\mathcal{R}_\lambda(\beta; F) = \mathbb{E}_{P_\pi}[\ell(D_\beta(W), Z)] + \lambda \|\beta\|_2^2, \quad \beta_\lambda(F) := \arg \min_{\beta} \mathcal{R}_\lambda(\beta; F), \quad (22)$$

with $\ell(D, z) = -z \log D - (1 - z) \log(1 - D)$ the logistic loss and $\lambda \geq 0$. Write $G(\beta, F) := \nabla_\beta \mathcal{R}_\lambda(\beta; F)$, so that an interior minimizer solves

$$G(\beta_\lambda(F), F) = 0, \quad (23)$$

and $H_\lambda := \nabla_\beta^2 \mathcal{R}_\lambda(\beta_\lambda; F)$ for the Hessian at that point.

Assumption 2 (Regularity). $\lambda > 0$; $\beta_\lambda(F)$ is an interior stationary point of $\mathcal{R}_\lambda(\cdot; F)$; F_β is twice continuously differentiable in β in a neighbourhood of $\beta_\lambda(F)$; the Hessian H_λ is invertible; and D_{β_λ} is bounded away from 0 and 1 on the support of P_π .

Assumption 2 is what the implicit function theorem needs, and no more. The two discriminants below differ in exactly one respect: whether it is a consequence of the specification or a hypothesis about the fitted point. For an affine logit it is a proposition (Section 5); for a network it must be posited (Section 6).

The chain rule for the logistic loss gives $\nabla_\beta \ell(D_\beta(w), z) = (D_\beta(w) - z)J_\beta(w)$, so that with

$$u_\beta(w) := (D_\beta(w) - 1)J_\beta(w), \quad v_\beta(w) := D_\beta(w)J_\beta(w) \quad (24)$$

the gradient of the loss at labels 1 and 0 is u_β and v_β respectively. Differentiating the first-order condition (23) along the contamination path (6) gives the central result of this section.

Theorem 1 (Influence function of the contrast estimate). *Under Assumption 2, with $\beta = \beta_\lambda(F)$ and $w = (t_w, x_w, y_w)$,*

$$\text{IF}(w; \beta_\lambda, F) = -H_\lambda^{-1} S(w; \beta), \quad S = S_1 + S_0^{TX} + S_0^Y, \quad (25)$$

where

$$\begin{aligned} S_1 &= \pi(u_\beta(w) - \mathbb{E}_{P_1}[u_\beta]), \\ S_0^{TX} &= (1 - \pi)(\mathbb{E}_{\tilde{Y} \sim F_Y}[v_\beta(t_w, x_w, \tilde{Y})] - \mathbb{E}_{P_0}[v_\beta]), \\ S_0^Y &= (1 - \pi)(\mathbb{E}_{(T, X) \sim F_{T, X}}[v_\beta(T, X, y_w)] - \mathbb{E}_{P_0}[v_\beta]). \end{aligned} \quad (26)$$

The three-channel form is the point of departure from the classical sandwich formula. The proof, in Appendices D.2 and D.3, applies the implicit function theorem to (23) and then differentiates the mixture law channel by channel. The interpretation follows the source of each channel. The first, S_1 , is the ordinary score of an M -estimator, computed under the law of the observed data; it is the only channel a fixed training distribution would produce. The other two arise entirely from the contrastive law: since $P_0(F)$ is a product of two marginals of F , contaminating F contaminates each factor separately, and the derivative of the product contributes one term per factor. These two channels are what the formulas of Avella-Medina (2017) and Koh and Liang (2017) lack. They also disappear when the negative class is drawn from a distribution fixed by the analyst, as in noise-contrastive estimation (Gutmann and Hyvärinen, 2010); in that case (25) collapses to the classical sandwich formula, as verified in Appendix D.4.

Nothing in (26) refers to the parameterization of the logit: the score is a property of the contrastive law alone. The Hessian is where the parameterization enters. Differentiating $(D_\beta - z)J_\beta$ once more in β gives

$$\nabla_\beta [(D_\beta - z)J_\beta] = D_\beta(1 - D_\beta)J_\beta J_\beta^\top + (D_\beta - z)\nabla_\beta^2 F_\beta, \quad (27)$$

and averaging over P_π splits the Hessian into a Gauss–Newton part and a curvature part,

$$H_\lambda = H_\lambda^{\text{GN}} + H_\lambda^{\text{curv}}, \quad \begin{cases} H_\lambda^{\text{GN}} = \mathbb{E}_{P_\pi}[D_\beta(1 - D_\beta)J_\beta J_\beta^\top] + 2\lambda I, \\ H_\lambda^{\text{curv}} = \mathbb{E}_{P_\pi}[(D_\beta - Z)\nabla_\beta^2 F_\beta], \end{cases} \quad (28)$$

the generalized Gauss–Newton split of the Hessian of a model with a sigmoid link and a loss convex in D (Martens, 2020). The first part is positive definite for free; the second carries no sign and vanishes identically when F_β is affine in β . This single display is the whole difference between the two discriminants below, and Section 6 is largely an account of its consequences.

4.6 Closed form of the three channels

Evaluating S_0^{TX} and S_0^Y appears to require two nested expectations for every w . It does not, and the reason is the same for every differentiable discriminant: because $\nabla_\beta D_\beta = D_\beta(1 - D_\beta)J_\beta$, the factor $D_\beta(1 - D_\beta)$ cancels against the derivative of the logistic loss, and each channel reduces to a plain expectation of the already-computed quantity $D_\beta J_\beta$. Define the three averaged Jacobians

$$\begin{aligned} b_{TX}(t_w, x_w) &:= \mathbb{E}_{\tilde{Y} \sim f_Y} [D_\beta J_\beta(t_w, x_w, \tilde{Y})], & b_Y(y_w) &:= \mathbb{E}_{(T, X) \sim f_{T, X}} [D_\beta J_\beta(T, X, y_w)], \\ b_0 &:= \mathbb{E}_{P_0} [D_\beta J_\beta]. \end{aligned} \quad (29)$$

Proposition 5 (Closed form). *Under Assumption 2,*

$$S(w; \beta) = \pi(D_\beta(w) - 1)J_\beta(w) + (1 - \pi)(b_{TX}(t_w, x_w) + b_Y(y_w) - b_0) + 2\lambda\beta_\lambda. \quad (30)$$

The derivation is in Appendix D.4. Each of b_{TX} and b_Y is a Monte Carlo average over draws resampled from a marginal, with no differentiation through the resampling step, which is what makes (30) computable at all. One feature of it is easy to get wrong in an implementation, and the next lemma guards against it: the three channels produce two contrastive centring terms, but the first-order condition (23) eliminates only one of them, so a single uncanceled term $-(1 - \pi)b_0$ survives.

Lemma 1 (Zero mean). $\mathbb{E}_{W \sim F} [S(W; \beta_\lambda)] = 0$, and hence $\mathbb{E}_{W \sim F} [\text{IF}(W; \beta_\lambda, F)] = 0$.

Appendix D.5 proves this using the fact that when $W \sim F$ the pair (T, X) follows $F_{T, X}$ and the outcome follows F_Y , so each contrastive channel averages to b_0 and the residual is exactly what makes the total vanish. Dropping the term $-(1 - \pi)b_0$ would leave a nonzero mean, which no influence function can have; the lemma therefore also serves as a numerical check on any implementation.

4.7 The correction in closed form

We can now return to the correction term left general in (19) and put it in closed form. That term needs $\text{IF}(w; D_\pi, F)$, the influence function of the fitted classifier evaluated pointwise at (t, X, y) , whereas Theorem 1 supplies $\text{IF}(w; \beta_\lambda, F)$, the influence function of the parameter: a single vector. The link between them is what makes the closed form possible. Once D_π is modelled as $D_{\beta_\lambda(F)}$, only $\beta_\lambda(F)$ is estimated from data, and the map $\beta \mapsto D_\beta(t, x, y)$, at any fixed (t, x, y) , is a known, non-random function, exactly like h in Lemma 5. The chain rule for influence functions therefore gives

$$\text{IF}(w; D_\pi(t, x, y), F) = \nabla_\beta D_\beta(t, x, y)^\top \text{IF}(w; \beta_\lambda, F). \quad (31)$$

Because $\text{IF}(w; \beta_\lambda, F)$ does not depend on (t, x, y) , and so not on the covariates being averaged over, it factors out of the expectation in the correction term of (19). That is what collapses it to an inner product, and it is why the algorithm only ever has to estimate $\text{IF}(w; \beta_\lambda, F)$ once per observation, never $\text{IF}(w; D_\pi(t, x, y), F)$ separately at every evaluation point.

Corollary 1. Let $a_\lambda(t, y) := \mathbb{E}_X [R(t, X, y) J_\beta(t, X, y)]$ and $\xi(t, y) := H_\lambda^{-1} a_\lambda(t, y)$. Then

$$\frac{1 - \pi}{\pi} \mathbb{E}_X \left[\frac{\text{IF}(w; D_\pi(t, X, y), F)}{(1 - D_\pi(t, X, y))^2} \right] = a_\lambda(t, y)^\top \text{IF}(w; \beta_\lambda, F) = -\xi(t, y)^\top S(w; \beta). \quad (32)$$

Appendix D.6 proves it. Combining Proposition 4, Theorem 1 and Corollary 1 turns the general estimator (19) into its closed form,

$$\hat{p}_t^{\text{ifc}}(y) = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{f}_Y(y) \left[\hat{R}(t, X_i, y) - \hat{m}(t, y) - \hat{\xi}(t, y)^\top \hat{S}(W_i) \right] + \hat{m}(t, y) K_{h_Y}(Y_i - y) \right\}, \quad (33)$$

with $\hat{m}(t, y) = n^{-1} \sum_j \hat{R}(t, X_j, y)$. This is the closed form promised in Section 4.3: the correction term of (19) has become the single inner product $\hat{\xi}(t, y)^\top \hat{S}(W_i)$, costing one linear solve per evaluation point and one inner product per observation. It uses no generalized propensity score and smooths only in the outcome coordinate.

4.8 The estimation procedure

Algorithm 1 states the whole procedure. It is complete for any discriminant satisfying Assumption 2: no step is left open, and Sections 5 and 6 change nothing in it beyond how the two linear-algebra steps, forming $\hat{H}_\lambda$ and solving against it, are carried out. Cross-fitting (Chernozhukov et al., 2018) runs throughout, the contrast function being fitted on one fold and the influence function evaluated on the other, with the roles exchanged and the two estimates averaged; this removes the empirical-process conditions that would otherwise be needed and makes the conditional-bias identity (5) hold exactly.

Algorithm 1 The IF-corrected estimator (any differentiable discriminant)

- 1: Split $\mathcal{D}_{\text{obs}}^{T,X,Y}$ into $\mathcal{D}_{\text{fit}}^{T,X,Y}$ and $\mathcal{D}_{\text{eval}}^{T,X,Y}$; exchange the roles at the end and average the two estimates.
 - 2: On $\mathcal{D}_{\text{fit}}^{T,X,Y}$, build $\mathcal{D}_{\text{ctr}}^{W,Z}$ satisfying MDCond(π) by labelling each observed vector $Z = 1$ and adjoining κ vectors with resampled outcomes labelled $Z = 0$, so that $\pi = 1/(1 + \kappa)$.
 - 3: Fit $\hat{\beta}_\lambda$ on $\mathcal{D}_{\text{ctr}}^{W,Z}$ by minimizing (22); set $\hat{D}_{\text{MCD}} = \sigma(F_{\hat{\beta}_\lambda})$ and $\hat{R} = \frac{1-\pi}{\pi} \hat{D}_{\text{MCD}} / (1 - \hat{D}_{\text{MCD}})$.
 - 4: Fit $\hat{f}_Y$ from the original outcomes in $\mathcal{D}_{\text{fit}}^{T,X,Y}$.
 - 5: Form $\hat{H}_\lambda$ from (28) on $\mathcal{D}_{\text{ctr}}^{W,Z}$ and prepare a solver for it; form $\hat{b}_0$ as an average of $D_\beta J_\beta$ over the contrastive part of $\mathcal{D}_{\text{ctr}}^{W,Z}$.
 - 6: **for** each W_i in $\mathcal{D}_{\text{eval}}^{T,X,Y}$ **do**
 - 7: compute $\hat{b}_{TX}(T_i, X_i)$ and $\hat{b}_Y(Y_i)$ by subsampling the corresponding marginals over S_{mc} draws;
 - 8: assemble $\hat{S}(W_i)$ from (30).
 - 9: **end for**
 - 10: **for** each evaluation point (t, y) **do**
 - 11: $\hat{m}(t, y) \leftarrow n^{-1} \sum_j \hat{R}(t, X_j, y)$;
 - 12: $\hat{a}_\lambda(t, y) \leftarrow n^{-1} \sum_j \hat{R}(t, X_j, y) J_{\hat{\beta}_\lambda}(t, X_j, y)$; solve $\hat{H}_\lambda \hat{\xi}(t, y) = \hat{a}_\lambda(t, y)$.
 - 13: **end for**
 - 14: **return** $\hat{p}_t^{\text{ifc}}(y)$ assembled from (33).
-

Two costs are worth reading off the algorithm, because they are what the next two sections trade against each other. Step 5 is paid once, not once per observation: the Hessian is a single average over $\mathcal{D}_{\text{ctr}}^{W,Z}$, prepared once and reused at every evaluation point. Step 7 is the inner Monte Carlo, and subsampling it over S_{mc} draws brings the loop to S_{mc} evaluations of $D_\beta J_\beta$ per observation, at the price of an $\mathcal{O}(S^{-1/2})$ error inside a term that is itself averaged over the sample. The one genuinely open question is step 5's solver, and the answer depends on whether H_λ^{curv} in (28) vanishes.

5 The penalized logistic discriminant

The first application of Section 4.5 takes F_β affine in β . There are two reasons for starting here. It is the loss and the parameterization under which Mezzani et al. (2025) establish the Bayes optimality of the contrast function, so it is the natural case to correct; and it is the one case in which Assumption 2 is not an assumption at all but a consequence of the specification. Nothing else in this section is new: Theorem 1, Proposition 5, Lemma 1 and Corollary 1 apply verbatim, and Algorithm 1 runs unchanged. What follows records only what is particular to this choice.

5.1 Specification

Let $\phi : \mathcal{W} \rightarrow \mathbb{R}^k$ be a fixed feature map and set

$$F_\beta(w) = \beta^\top \phi(w), \quad \text{so that} \quad J_\beta(w) = \phi(w) \quad \text{and} \quad \nabla_\beta^2 F_\beta \equiv 0. \quad (34)$$

The Jacobian is the feature map itself, and it does not depend on β .

Assumption 3 (Regularity, logistic discriminant). $\lambda > 0$, the feature map ϕ is bounded on $\mathcal{W}$, and D_{β_λ} is bounded away from 0 and 1 on the support of P_π .

Proposition 6 (Assumption 2 holds, and globally). *Under Assumption 3, the criterion (22) is strictly convex, so $\beta_\lambda(F)$ exists, is unique and is interior; $H_\lambda^{\text{curv}} = 0$, so $H_\lambda = H_\lambda^{\text{GN}}$; and every eigenvalue of H_λ is at least 2λ , at every β and not only at $\beta_\lambda(F)$.*

Appendix D.1 proves it, and the proof is short: with $\nabla_\beta^2 F_\beta \equiv 0$ the curvature term of (28) vanishes identically, leaving a positive-semidefinite Gram matrix plus $2\lambda I$. The unpenalized case $\lambda = 0$ remains admissible provided ϕ is not collinear on the support. This is the distinction that Section 6 loses: Assumption 3 constrains the *specification*, and invertibility follows as a theorem, uniformly in β .

5.2 What the general results become

Substituting $J_\beta = \phi$ into Section 4.5 gives the estimator in fully explicit form. The score (30) reads

$$S(w; \beta) = \pi(D_\beta(w) - 1)\phi(w) + (1 - \pi)\left(b_{TX}(t_w, x_w) + b_Y(y_w) - b_0\right) + 2\lambda\beta_\lambda, \quad (35)$$

with the three averaged feature vectors of (29) taken at $J_\beta = \phi$, and the collapse vector of Corollary 1 is $a_\lambda(t, y) = \mathbb{E}_X[\mathbf{R}(t, X, y)\phi(t, X, y)]$. Three consequences are worth naming, because they are exactly what the neural-network case gives up.

- *The Hessian is small, fixed and factorizable.* It is $k \times k$ with k chosen by the analyst, it does not move with β beyond the scalar weights $D_\beta(1 - D_\beta)$, and step 5 of Algorithm 1 can form it explicitly and take a single Cholesky factorization per fold. Every subsequent solve $\hat{H}_\lambda \hat{\xi}(t, y) = \hat{a}_\lambda(t, y)$ is then a triangular solve rather than a fresh $\mathcal{O}(k^3)$ inversion.
- *The inner Monte Carlo is over a fixed map.* Since ϕ does not depend on β , b_{TX} and b_Y are averages of $D_\beta \phi$ over resampled draws with no backward pass anywhere in the loop, bringing step 7 to $\mathcal{O}(nS_{\text{mc}}k)$ arithmetic.
- *No local hypothesis is carried.* By Proposition 6 the linear system of Corollary 1 is solvable by construction, with an explicit lower bound 2λ on the spectrum, so nothing needs to be checked at run time.

5.3 Implementation

Remark 1 (Implementation). Three details of Algorithm 1, left generic above, are fixed to specific choices in the implementation used for Section 7, and are worth recording here rather than only in code. First, $\widehat{H}_\lambda$ (step 5) is factorized once per fold by a Cholesky decomposition, as just described. Second, λ defaults to 0.5 regardless of k : a value an order of magnitude larger than what bare well-specification arguments would suggest, adopted after direct measurement (Appendix G) showed that a small λ leaves $\widehat{\beta}_\lambda$ overconfident whenever ϕ is even mildly misspecified for the data-generating process – harmless for the plug-in estimate, which only needs $\widehat{R}$ to be reasonable, but propagated by (30) directly into $\widehat{S}$ and hence into the correction term. Third, the Monte Carlo average defining $\widehat{b}_{TX}, \widehat{b}_Y$ (step 7) is computed in batches capped so that no single batch exceeds a fixed number of scalar entries, independently of k ; this keeps memory use bounded on a GPU as k grows into the thousands, at no change to the value being estimated, only to how it is accumulated. All three choices are orthogonal to the theory of Sections 4.5–5.2, which holds for any $\lambda > 0$ and any exact evaluation of the same sums; they matter only for the wall-clock and memory cost measured in Appendix H.

5.4 Asymptotics

[TO COMPLETE: À voir.]

6 The neural-network discriminant

The second application takes F_β nonlinear in β . Again, nothing in Section 4.5 needs restating: Theorem 1, Proposition 5, Lemma 1 and Corollary 1 hold as written, with J_β now the network’s logit Jacobian, and the closed-form estimator (33) is unchanged. What this section records is the one thing that does change, the Hessian, and the two consequences it has: a regularity condition that must be assumed rather than proved, and a linear solve that can no longer be a factorization.

6.1 Specification

Let $F_\beta : \mathcal{W} \rightarrow \mathbb{R}$ be a multilayer perceptron of fixed depth and width, β collecting its P weights, and $D_\beta(w) = \sigma(F_\beta(w))$. Then

$$J_\beta(w) = \nabla_\beta F_\beta(w) \in \mathbb{R}^P, \quad \nabla_\beta^2 F_\beta \not\equiv 0, \quad (36)$$

the Jacobian computed by one backward pass. J_β plays the role ϕ played in Section 5.1, with the one difference that drives everything below: ϕ was fixed, and J_β depends on β .

Assumption 4 (Local regularity). Assumption 2 holds at the fitted stationary point $\beta_\lambda(F)$; in particular H_λ is invertible there.

Assumption 4 has the same content as Assumption 3 and a different status, and the difference is the whole story of this section. Assumption 3 constrains the specification, and Proposition 6 then *derives* invertibility at every β . Assumption 4 instead posits it directly, *at the one point $\beta_\lambda(F)$ that matters*, because nothing about a multilayer F_β implies it.

6.2 Why it must be assumed

The reason is visible in the Hessian split (28), which for a network retains both terms.

Proposition 7 (Gauss–Newton decomposition). $H_\lambda^{\text{GN}} \succeq 2\lambda I \succ 0$ always. $H_\lambda^{\text{curv}} = 0$ whenever F_β is affine in β , and is sign-indefinite in general.

H_λ^{GN} is literally the Hessian of Appendix D.1 with J_β in place of ϕ , and is invertible for the same reason, at every β . What is new, and cannot be argued away, is H_λ^{curv} : it is generically nonzero for a multilayer F_β , it carries no sign, and no argument available here rules out its cancelling part or all of H_λ^{GN} ’s positivity at the specific point $\beta_\lambda(F)$. That is exactly the content of Assumption 4. It is small when $D_\beta(w)$ is close to 0 or 1 on P_1 and close to 0 on P_0 , that is, when the classifier already separates real from recombined vectors well: a heuristic in favour of the assumption being reasonable in practice, not a proof that it holds.

6.3 Solving against the Hessian

Two changes to step 5 of Algorithm 1 follow, and nothing else in it moves. First, H_λ is $P \times P$ with P the number of network weights, ordinarily far too large to form or factorize; every linear system in it is instead solved by an iterative method using Hessian-vector products alone, each costing one extra backward pass (the Pearlmutter trick) and never requiring $\hat{H}_\lambda$ to be written down, as already used for this purpose by Koh and Liang (2017). The Cholesky factorization of Section 5.2 is not available, so the solve is paid once per evaluation point rather than amortized over all of them.

Second, which matrix to solve against is a modelling choice, not an implementation detail. Proposition 7 guarantees $\hat{H}_\lambda^{\text{GN}}$ positive definite, but not $\hat{H}_\lambda$ itself, and the two coincide only when $\hat{H}_\lambda^{\text{curv}} \approx 0$. The recommended default is the *exact* Hessian-vector product, no harder to compute than the Gauss–Newton one, still one Pearlmutter pass, solved by an indefinite-tolerant method (e.g. MINRES, or conjugate gradients with a small damping term if $\hat{H}_\lambda$ is found indefinite); this targets the linear system Theorem 1 actually specifies, with no approximation. Substituting $\hat{H}_\lambda^{\text{GN}}$ remains available as a cheaper fallback, always positive definite by construction, but its accuracy is regime-dependent rather than automatic: Proposition 7 ties it to the classifier already separating real from recombined vectors well, and it can be a poor substitute otherwise, for instance for a small, under-trained network with large residuals $D_\beta - z$ on P_1 , where H_λ^{curv} is exactly what is large.

One step has no counterpart in Algorithm 1 and must be inserted after the fit: a numerical check of Assumption 4. Estimate the smallest eigenvalues of $\hat{H}_\lambda$ by Hessian-vector products and Lanczos, to detect near-singularity or indefiniteness, and compute the ratio $\|\hat{H}_\lambda^{\text{curv}}\|/\|\hat{H}_\lambda^{\text{GN}}\|$, which is what decides whether the cheaper Gauss–Newton solve is trustworthy. Note also that every $J_{\hat{\beta}_\lambda}$ in steps 7 and 12 of Algorithm 1 now costs a backward pass, one per Monte Carlo draw, where the logistic case only read off a fixed ϕ .

A direct empirical check of this trade-off belongs in Section 7: compare the Gauss–Newton and exact-Hessian estimates of $\text{IF}(w; \beta_\lambda, F)$ against a finite-difference derivative of $\hat{\beta}_\lambda(F_\epsilon)$ along the contamination path (6), as a function of the same ratio. Doing so turns Assumption 4 from an unverifiable hypothesis into a diagnostic with a quantitative threshold, in the same spirit as the finite-population check used to validate Theorem 1 in the logistic case.

7 Simulation study

[TO COMPLETE: This section states the experimental protocol in full; results, figures and tables are added once the runs described below, on the compute environment of Appendix H, with hyperparameters selected by Appendix G, complete.]

7.1 Validating the influence function

Before assembling the estimator, we check that (25) really is the influence function it claims to be: a purely numerical statement, independent of any asymptotic argument, and worth verifying on its own before any sampling variability is allowed to obscure a possible error in the formula itself.

We build a finite population F of fixed size (feature map ϕ and penalty λ fixed in advance), for which $\beta_\lambda(F)$ is the exact, unique minimizer of the finite-sum criterion (22). For a probe point w and a grid of $\varepsilon \rightarrow 0$, F is perturbed along the contamination path $F_\varepsilon = (1 - \varepsilon)F + \varepsilon\delta_w$ of (6), and (22) is re-solved exactly at every ε to form the centred finite difference of $\beta_\lambda(F_\varepsilon)$. This finite difference is the ground truth against which three predictions are compared, in relative L_2 norm and cosine similarity, as functions of ε : the closed-form three-channel formula (30) of Theorem 1; the single-channel “classical sandwich” formula of Avella-Medina (2017); Koh and Liang (2017), which keeps only S_1 and drops S_0^{TX}, S_0^Y , included as a negative control that must fail once the two contrastive channels are not negligible; and the mean-zero identity of Lemma 1, tracked as $\varepsilon \rightarrow 0$ as a further consistency check. The same three comparisons are repeated for the neural-network discriminant, where Theorem 1 holds with J_β in place of ϕ , with the exact-Hessian and Gauss–Newton solves of Proposition 7 evaluated separately, as functions of $\|\hat{H}_\lambda^{\text{curv}}\|/\|\hat{H}_\lambda^{\text{GN}}\|$; this turns Assumption 4, otherwise unverifiable, into a diagnostic with a quantitative threshold (Section 6.3).

A second layer of the same check moves from the population to a sample: Algorithm 1 is run on samples of size n drawn from F , over the (n, S_{mc}) grid of Appendix G, S_{mc} the Monte Carlo size used for $\hat{b}_{TX}, \hat{b}_Y$, and its output is compared to the population influence function computed above. An ablation recomputes $\hat{b}_{TX}, \hat{b}_Y$ *exactly* over the sample’s atoms rather than by Monte Carlo subsampling, which isolates the $\mathcal{O}(S^{-1/2})$ error contributed by subsampling from the $\mathcal{O}(n^{-1/2})$ error already present in a finite sample of size n . Throughout, the empirical mean of $\hat{S}(W_i)$ is tracked against n and S_{mc} , as a running check of Lemma 1 that any correct implementation must pass regardless of the outcome of the comparisons above.

7.2 Data-generating processes

Four families of data-generating processes are used, each for a different purpose; Table 2 previews all four before the details below.

Family	Treatment	What is swept	Purpose
AXY (App. F.1)	binary	cov_dim(X)	binary treatment vs. confounder dimension; classical vs. IF-corrected cross-check (<i>e</i> closed form)
Linear-Gaussian (App. F.2)	continuous natively, binary as run	cov_dim(Z)	well-specified control: the estimation-only regime, with no approximation error
Four continuous models (App. F.3)	continuous, dim(T)	dim(X), dim(T)	distributional shapes; the d -crossover of Section 8.2
Real / semi-synthetic (App. F.4)	binary	,	companion chapter’s broader comparison

Table 2: The data-generating families of Section 7.2.

AXY (Appendix F.1), adapted from the toy structural model of Melnychuk et al. (2023), has a binary treatment, a closed-form e , and a covariate dimension cov_dim(X) that can be swept into

the thousands, the binary-treatment analogue of the d -sweep below. The closed-form e is exactly the ingredient needed to compute the classical estimator (9) directly and check that it agrees with the IF-corrected one (33) up to sampling error (Section 7.3); Figure 2 shows its outcome mechanism. This degenerate, binary-treatment case is the smallest instance, inside this paper, of a broader observation: on the simulated AXY design and on real data, the companion chapter’s comparison against the baselines of Appendix J finds the MCD-based estimator competitive even outside the continuous, multidimensional regime it targets.

The **well-specified linear-Gaussian model** (Appendix F.2) is a control: a Gaussian confounder and a linear-Gaussian outcome, for which the contrast function’s log-odds is an exact quadratic polynomial in (t, x, y) , so a degree-two feature map is not an approximation but an exact representation. AXY and the four models below are all, by design, misspecified for such a feature map; this one is not, and is what isolates the estimation-only regime, in which the parametric family contains the contrast function exactly and nothing is left for an approximation error to contribute, from the approximation-plus-estimation regime the other designs exercise.

The **four continuous models** (Appendix F.3) share the paper’s actual regime, confounding jointly Gaussian in (X, T) , both dimensions free to vary, and differ only in the outcome law $Y | T, X$, each adding one property a jointly Gaussian (X, T, Y) cannot produce: a mixing weight that depends on t (multimodal), a Student- t noise term (heavy tails), a variance that grows with t (heteroscedastic), or a skew-normal noise term (skewed). Figure 3 shows all four. Every one keeps an elementary or one-dimensional-quadrature closed form for $p_t(y)$, which is what makes it usable as ground truth rather than only a qualitative illustration; since $\text{do}(T = t)$ severs $X \rightarrow T$, only the outcome mechanism, never the confounding mechanism, can shape $p_t(y)$, which is why varying it is enough.

Finally, the **real and semi-synthetic datasets** (IHDP, ACIC 2016/2018, HC-MNIST; Appendix F.4) that Melnychuk et al. (2023) compare their baselines on all have a binary treatment; they belong to the companion, more applied chapter’s broader comparison, not to this simulation study, which stays confined to the IF-corrected estimator and its two discriminants.

7.3 Experiments

For each data-generating process, the corrected estimator (33) and, where feasible, the classical estimator (9) are compared on bias and root mean squared error (RMSE) against the known $p_t(y)$, as functions of the sample size n ; on AXY, where e has a closed form, the two estimates are additionally checked directly against one another, which they should agree with up to sampling error.

Two dimensions are then swept independently, to separate the two competing effects behind the approximation/estimation trade-off: the feature-map dimension k , to trace the approximation/estimation trade-off, and the treatment dimension $d = \dim(T)$, to locate the crossover discussed in Section 8.2, where the contrastive route’s advantage over the classical, kernel-based one is expected to grow. On the four continuous data-generating processes, the two discriminants of Sections 5 and 6 are further compared against one another on accuracy and wall-clock cost, and each process’s own difficulty parameter, component separation for the mixture, degrees of freedom ν for the Student- t , the variance’s growth rate for the heteroscedastic case, skewness for the last one, is varied at fixed n and d , to see how quickly each discriminant’s accuracy degrades as the target moves away from what a linear-in-features model can represent well.

Every configuration is run for 50 independent replications, each an independent draw of the entire

data-generating process from a fresh seed, not a resampling of one fixed dataset — so that the resulting variance is the Monte Carlo variance of the estimator, not resampling variance around a single realization. Every replication’s full result (bias, RMSE, wall-clock time, the fitted hyperparameters, and the seed that produced it) is stored individually rather than only its average, so that quantiles, violin plots, and a true-vs-predicted density plot for any single stored replication can all be produced after the fact, from the seed alone, without rerunning the whole configuration. Hyperparameters, λ , the Monte Carlo size S_{mc} , and each discriminant’s own parameters, are selected once per configuration by the cross-validation protocol of Appendix G and then held fixed across all 50 replications, mirroring the tuning protocol of Melnychuk et al. (2023) rather than re-tuning at every replication. The compute environment, and the policy governing when a run uses the GPU, are described in Appendix H.

8 Discussion

Having built and analyzed the IF-corrected estimator, we now step back to say what its guarantee is and is not, how to check it in practice, and how it sits in the surrounding literature.

8.1 Scope of the guarantee

The two estimators of this paper differ in a way that is easy to state and worth understanding: the classical estimator (9) is doubly robust in the pair $(e, f_{Y|T,X})$, whereas the IF-corrected estimator (33) is not doubly robust. The difference is structural rather than an artefact of the derivation, and two facts explain it.

The classical pair consists of the two legs of the identified functional, the treatment law and the outcome law, and double robustness is the statement that an error in one leg is harmless if the other is estimated well. The contrastive factorization (13) dismantles this arrangement: it removes the treatment leg from the formula and splits the outcome leg into a marginal and a density ratio. What takes the place of the pair is a single fitted object, the contrast function, and the second-order remainder of the IF-corrected estimator is accordingly the *square* of one nuisance error, $\|\hat{\beta} - \beta_\lambda\|^2$, rather than a product of two errors that could cancel. No second leg remains whose good behaviour could rescue a poor contrast fit.

The two factors are moreover not variationally independent, so one could not treat them as an error-cancelling pair even in principle.

Proposition 8 (Normalization constraint). *For every y with $f_Y(y) > 0$,*

$$\mathbb{E}_{(T,X)}[R(T, X, y)] = 1. \tag{37}$$

Appendix E.1 proves this. The constraint holds for every y , so the contrast function is tied to the marginal through a continuum of identities, and the two are not free to vary independently.

What the correction does provide is the removal of the first-order bias (5) — which is present regardless of any pairing of nuisances, together with first-order insensitivity to the estimation error of β . The condition that replaces double robustness bears on a single object, the approximation error of the contrast model.

8.2 Nature of the simplification

The contrastive route does not remove work; it changes its kind.

What it removes is two objects that are hard for the same reason. The classical estimator (9) needs $\hat{e}(t \mid X_i)$, a conditional density on $\mathbb{R}^d$, and smooths it with a d -dimensional kernel K_{h_T} ; Appendix B.1 requires $nh_T^d h_Y \rightarrow \infty$ for this to be feasible at all, and it is exactly this condition that a vector treatment strains. The contrastive estimator (33) needs neither: it smooths only in y , and its estimation error is governed by $\|\hat{\beta} - \beta_\lambda\|^2$, a parametric rate for a fixed-dimensional β . Removing the treatment-coordinate kernel is at least as much of the gain as removing e by name; the two are the same difficulty, since e is precisely what made that kernel necessary.

What it adds: computing $\text{IF}(w; D_\pi, F)$ requires the three-channel score (26), a Hessian H_λ , and, in closed form, one linear solve per evaluation point (Corollary 1). Weighed purely in operations, this is not obviously cheaper than fitting $\hat{e}$; what makes it cheaper is that it is a computation over a k -dimensional object rather than an estimation problem on $\mathbb{R}^d$, a difference in kind, not just in constant.

That the added machinery turns out to be cheap is a property of the logistic discriminant, not of the contrastive route in general. It is cheap here because $S(w; \beta)$ has the closed form (30), H_λ is a single average factorized once, and b_{TX}, b_Y are plain Monte Carlo averages of an already-computed quantity (Section 5.2). None of this is automatic: for the neural-network discriminant H_λ can fail to be invertible at all, and the tractable substitute requires conjugate-gradient solves and an unverifiable local regularity assumption (Section 6.2). Penalized logistic regression is the instance for which the trade is unambiguously favourable.

Two qualifications remain even in that favourable case. First, the double robustness given up above matters: with the treatment leg gone, nothing rescues the estimator if ϕ is misspecified, and the correction’s validity rests on the single quantity $\|D_{\beta_\lambda} - D_\pi\|_{L_2}$, where the classical route has two independent chances to be right. Second, the case for the trade grows with d . At $d = 1$, with e itself simple to estimate, the classical route needs no high-dimensional kernel to begin with, keeps double robustness, and the comparison may favour it. The contrastive route’s advantage is specific to the high-dimensional, vector-treatment regime this paper targets, not universal.

9 Conclusion

[TO COMPLETE: To complete.]

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Appendix

The appendix collects all derivations, and each is written to be self-contained. Appendix A states the rules of the perturbation calculus used throughout. Appendix B derives the influence function of the interventional density and the remainder of the associated estimator, proving Propositions 1 and 2. Appendix C derives the contrastive form of the correction, establishing Propositions 3 and 4. Appendix D treats the differentiable contrast, proving Theorem 1, Proposition 5, Lemma 1, Corollary 1 and, for the affine logit, Proposition 6. Appendix E contains the normalization constraint (Proposition 8). The three appendices that follow support Section 7 rather than the theory above: Appendix F gives the full specification of every data-generating process used, Appendix G the cross-validation protocol and hyperparameter grids, and Appendix H the compute environment and the policy for when a run uses the GPU. Two reference tables close the appendix: Appendix I lists every abbreviation and every symbol used in the paper, and Appendix J the baseline estimators of Section 1.

A Perturbation calculus

Throughout this appendix $F_\varepsilon = (1 - \varepsilon)F + \varepsilon\delta_w$ is the contamination path of (6), and IF is the influence functional of Definition 3. The single idea behind every rule below is that $\varepsilon \mapsto \psi(F_\varepsilon)$ is a real-valued function of one real variable, so the ordinary rules of differentiation apply to it; each lemma is one such rule, translated into the notation of IF. We use these lemmas without further comment in Appendices B–E.

Lemma 2 (Linearity). *For functionals ψ_1, ψ_2 admitting influence functions at F and constants $a, b \in \mathbb{R}$,*

$$\text{IF}(w; a\psi_1 + b\psi_2, F) = a \text{IF}(w; \psi_1, F) + b \text{IF}(w; \psi_2, F).$$

Proof. The map $\varepsilon \mapsto a\psi_1(F_\varepsilon) + b\psi_2(F_\varepsilon)$ is a linear combination of two differentiable functions, so its derivative at $\varepsilon = 0$ is the same linear combination of the two derivatives. By Definition 3 those derivatives are $\text{IF}(w; \psi_1, F)$ and $\text{IF}(w; \psi_2, F)$, which gives the claimed identity. $\square$

Lemma 3 (Product). $\text{IF}(w; \psi_1\psi_2, F) = \psi_1(F) \text{IF}(w; \psi_2, F) + \psi_2(F) \text{IF}(w; \psi_1, F)$.

Proof. Set $\varphi(\varepsilon) = \psi_1(F_\varepsilon)$ and $\chi(\varepsilon) = \psi_2(F_\varepsilon)$. The ordinary product rule gives $(\varphi\chi)'(0) = \varphi(0)\chi'(0) + \chi(0)\varphi'(0)$. Evaluating the four factors at $\varepsilon = 0$ using Definition 3, we have $\varphi(0) = \psi_1(F)$, $\chi(0) = \psi_2(F)$, $\varphi'(0) = \text{IF}(w; \psi_1, F)$ and $\chi'(0) = \text{IF}(w; \psi_2, F)$. Substituting yields $\psi_1(F) \text{IF}(w; \psi_2, F) + \psi_2(F) \text{IF}(w; \psi_1, F)$, as claimed. $\square$

Lemma 4 (Quotient). *If $\psi_2(F) \neq 0$,*

$$\text{IF}\left(w; \frac{\psi_1}{\psi_2}, F\right) = \frac{\text{IF}(w; \psi_1, F)}{\psi_2(F)} - \frac{\psi_1(F)}{\psi_2(F)^2} \text{IF}(w; \psi_2, F).$$

Proof. With φ, χ as in the previous proof and $\chi(0) = \psi_2(F) \neq 0$, the ordinary quotient rule gives $(\varphi/\chi)'(0) = \{\varphi'(0)\chi(0) - \varphi(0)\chi'(0)\}/\chi(0)^2$. Substituting the values of $\varphi(0), \chi(0), \varphi'(0), \chi'(0)$ from Definition 3 and separating the two fractions gives the stated expression. $\square$

Lemma 5 (Chain rule). *For $h : \mathbb{R} \rightarrow \mathbb{R}$ differentiable, $\text{IF}(w; h \circ \psi, F) = h'(\psi(F)) \text{IF}(w; \psi, F)$.*

Proof. With $\chi(\varepsilon) = \psi(F_\varepsilon)$, the ordinary chain rule gives $(h \circ \chi)'(0) = h'(\chi(0))\chi'(0)$. Since $\chi(0) = \psi(F)$ and $\chi'(0) = \text{IF}(w; \psi, F)$, this is $h'(\psi(F)) \text{IF}(w; \psi, F)$. $\square$

Lemma 6 (Odds transformation). *Let ψ take values in $(0, 1)$ and $h(u) = u/(1 - u)$. Then*

$$\text{IF}(w; h(\psi), F) = \frac{\text{IF}(w; \psi, F)}{(1 - \psi(F))^2}.$$

In particular, since $R = \frac{1-\pi}{\pi} h(D_\pi)$ with π constant,

$$\text{IF}(w; R, F) = \frac{1 - \pi}{\pi} \frac{\text{IF}(w; D_\pi, F)}{(1 - D_\pi)^2}. \quad (38)$$

Proof. Write $h(\psi) = \psi_1/\psi_2$ with $\psi_1 = \psi$ and $\psi_2 = 1 - \psi$. By Lemma 2, $\text{IF}(w; 1 - \psi, F) = -\text{IF}(w; \psi, F)$. Applying Lemma 4 and substituting this,

$$\text{IF}(w; h(\psi), F) = \frac{\text{IF}(w; \psi, F)}{1 - \psi(F)} + \frac{\psi(F)}{(1 - \psi(F))^2} \text{IF}(w; \psi, F) = \text{IF}(w; \psi, F) \frac{(1 - \psi(F)) + \psi(F)}{(1 - \psi(F))^2},$$

and the numerator equals 1, which gives the first display. For the second, apply Lemma 2 with the constant $\frac{1-\pi}{\pi}$ (whose influence function is zero, being law-independent) to $R = \frac{1-\pi}{\pi} h(D_\pi)$ and substitute $\psi = D_\pi$ into the first display; this is exactly (38). $\square$

Lemma 7 (Expectation of a law-dependent integrand). *Let $h(\cdot; F)$ depend on F both through the law of the integration variable and through its functional form. Then*

$$\text{IF}(w; \mathbb{E}_F[h(W; F)], F) = h(w; F) - \mathbb{E}_F[h(W; F)] + \mathbb{E}_F[\text{IF}(w; h(\cdot; F), F)]. \quad (39)$$

Proof. By the definition of F_ε as a mixture, expectation under F_ε splits as

$$\mathbb{E}_{F_\varepsilon}[h(W; F_\varepsilon)] = (1 - \varepsilon) \mathbb{E}_F[h(W; F_\varepsilon)] + \varepsilon h(w; F_\varepsilon).$$

Differentiate at $\varepsilon = 0$, term by term. In the first summand the product rule gives two pieces: differentiating the factor $(1 - \varepsilon)$ contributes $-\mathbb{E}_F[h(W; F)]$, and differentiating the integrand contributes $\mathbb{E}_F[\partial_\varepsilon h(W; F_\varepsilon)|_0] = \mathbb{E}_F[\text{IF}(w; h(\cdot; F), F)]$, the interchange of derivative and expectation being licensed by the regularity assumed where the lemma is invoked. In the second summand, the factor ε multiplies the derivative of $h(w; F_\varepsilon)$, which therefore drops out at $\varepsilon = 0$, while the derivative of ε times $h(w; F_\varepsilon)$ contributes $h(w; F)$. Summing the three surviving pieces gives (39). $\square$

The third term of (39) is what makes this lemma more than the standard formula for the influence function of an expectation: it accounts for the dependence of the integrand on the law, and is present precisely when the integrand is itself estimated, as the density ratio is in Section 4.

B The nonparametric correction

This appendix proves the results of Section 3. Appendix B.1 states the regularity conditions; Appendix B.2 derives the influence function of Proposition 1; Appendix B.3 regularizes its point masses and produces the estimator (9); and Appendix B.4 bounds the remainder, proving Proposition 2.

B.1 Regularity conditions

Proposition 1 holds under Assumption 1 together with the following conditions, which are those of Kennedy et al. (2017) and Kennedy et al. (2023) adapted to the present target. The joint density $f_{T,X,Y}$ is bounded and twice continuously differentiable in (t, y) in a neighbourhood of the evaluation point; the map $y \mapsto f_Y(y)$ is twice continuously differentiable at y ; and the kernels K_T, K_Y are symmetric, of second order, with compact support. The bandwidths satisfy $h_T, h_Y \rightarrow 0$, $nh_T^d h_Y \rightarrow \infty$, and the undersmoothing conditions $\sqrt{nh_T^d h_Y (h_T^2 + h_Y^2)} \rightarrow 0$. The smoothness in (t, y) is what allows the point masses to be replaced by kernels with a controlled bias in Appendix B.3; the undersmoothing conditions make that bias negligible on the $\sqrt{n}$ scale.

B.2 Derivation of the influence function

We prove Proposition 1. Write f_ε for the density of F_ε and express the target along the contamination path as

$$p_t(y)(F_\varepsilon) = \int_{\mathcal{X}} \frac{f_\varepsilon(t, x, y) f_\varepsilon(x)}{f_\varepsilon(t, x)} dx.$$

Because the contamination path is a mixture, $\varepsilon \mapsto f_\varepsilon$ is affine, and each density derivative at $\varepsilon = 0$ is the difference between the corresponding point mass and the density:

$$\begin{aligned} \partial_\varepsilon f_\varepsilon(t, x, y)|_0 &= \Delta(t, x, y) - f_{T,X,Y}(t, x, y), \\ \partial_\varepsilon f_\varepsilon(x)|_0 &= \Delta(x) - f_X(x), \\ \partial_\varepsilon f_\varepsilon(t, x)|_0 &= \Delta(t, x) - f_{T,X}(t, x), \end{aligned} \tag{40}$$

where $\Delta(t, x, y) := \delta_{t_w}(t) \delta_{x_w}(x) \delta_{y_w}(y)$ and analogously for the marginals. Writing the integrand as AB/C with $A = f_\varepsilon(t, x, y)$, $B = f_\varepsilon(x)$ and $C = f_\varepsilon(t, x)$, ordinary calculus gives

$$\partial_\varepsilon \left(\frac{AB}{C} \right) = \frac{\dot{A}B}{C} + \frac{A\dot{B}}{C} - \frac{AB\dot{C}}{C^2}.$$

Substituting (40) at $\varepsilon = 0$ and factoring the common quantity $f_{T,X,Y}(t, x, y) f_X(x) / f_{T,X}(t, x)$ out of the bracket,

$$\partial_\varepsilon p_t(y)(F_\varepsilon)|_0 = \int_{\mathcal{X}} \frac{f_{T,X,Y}(t, x, y) f_X(x)}{f_{T,X}(t, x)} \left[\frac{\Delta(t, x, y)}{f_{T,X,Y}(t, x, y)} + \frac{\Delta(x)}{f_X(x)} - \frac{\Delta(t, x)}{f_{T,X}(t, x)} - 1 \right] dx, \tag{41}$$

the constant -1 collecting the three density terms $-1 - 1 + 1$ of the three derivatives. We now evaluate the four contributions in turn.

The term in $\Delta(x)$ contains a point mass only in the integration variable x , so the integral collapses at $x = x_w$:

$$\int_{\mathcal{X}} \frac{f_{T,X,Y}(t, x, y)}{f_{T,X}(t, x)} \delta_{x_w}(x) dx = \frac{f_{T,X,Y}(t, x_w, y)}{f_{T,X}(t, x_w)} = f_{Y|T=t, X=x_w}(y) = \eta(x_w),$$

the integrand being continuous at x_w by Appendix B.1.

The term in $\Delta(t, x, y)$ carries point masses in t and y , which are *not* integration variables and therefore remain as Dirac factors; only $\delta_{x_w}(x)$ is integrated. Using $e(t | x) = f_{T,X}(t, x) / f_X(x)$,

$$\int_{\mathcal{X}} \frac{f_X(x)}{f_{T,X}(t, x)} \delta_{t_w}(t) \delta_{x_w}(x) \delta_{y_w}(y) dx = \delta_{t_w}(t) \delta_{y_w}(y) \frac{f_X(x_w)}{f_{T,X}(t, x_w)} = \frac{\delta_{t_w}(t) \delta_{y_w}(y)}{e(t | x_w)}.$$

The term in $\Delta(t, x)$ carries a point mass in t only, and integrating out $\delta_{x_w}(x)$ gives, again using the definition of e ,

$$-\int_{\mathcal{X}} \frac{f_{T,X,Y}(t, x, y) f_X(x)}{f_{T,X}(t, x)^2} \delta_{t_w}(t) \delta_{x_w}(x) dx = -\delta_{t_w}(t) \frac{f_{T,X,Y}(t, x_w, y)}{f_{T,X}(t, x_w)} \frac{f_X(x_w)}{f_{T,X}(t, x_w)} \cdot \frac{f_{T,X}(t, x_w)}{f_X(x_w)}$$

which simplifies to $-\delta_{t_w}(t) \eta(x_w)/e(t | x_w)$.

The constant term -1 integrates against the factored density to $-\mathbb{E}_X[\eta(X)] = -p_t(y)$ by Definition 1.

Collecting the four contributions,

$$\partial_\varepsilon p_t(y)(F_\varepsilon)|_0 = \frac{\delta_{t_w}(t) \delta_{y_w}(y)}{e(t | x_w)} - \frac{\delta_{t_w}(t) \eta(x_w)}{e(t | x_w)} + \eta(x_w) - p_t(y),$$

and factoring $\delta_{t_w}(t)/e(t | x_w)$ out of the first two terms yields

$$\text{IF}(w; p_t(y), F) = \frac{\delta_{t_w}(t)}{e(t | x_w)} (\delta_{y_w}(y) - \eta(x_w)) + \eta(x_w) - p_t(y),$$

which is exactly (8), the claim of Proposition 1. $\square$

B.3 Regularization of the point evaluations

Expression (8) is exact but not computable, because it contains Dirac masses in t and y . We regularize each by a kernel. Let $K_T : \mathbb{R}^d \rightarrow \mathbb{R}_+$ and $K_Y : \mathbb{R} \rightarrow \mathbb{R}_+$ be symmetric second-order kernels integrating to one, and for bandwidths $h_T, h_Y > 0$ set

$$K_{h_T}(u) = h_T^{-d} K_T(u/h_T), \quad K_{h_Y}(v) = h_Y^{-1} K_Y(v/h_Y). \quad (42)$$

Substituting $\delta_{t_w}(t) \rightsquigarrow K_{h_T}(t_w - t)$ and $\delta_{y_w}(y) \rightsquigarrow K_{h_Y}(y_w - y)$ in (8), and leaving the nuisances unchanged, gives the smoothed influence function

$$\text{IF}_h(w; p_t(y), F) = \frac{K_{h_T}(t_w - t)}{e(t | x_w)} (K_{h_Y}(y_w - y) - f_{Y|T=t, X=x_w}(y)) + f_{Y|T=t, X=x_w}(y) - p_t(y), \quad (43)$$

Write out the one-step construction (7) at $F = \hat{F}$: the plug-in $\hat{p}_t^{\text{pi}}(y) = n^{-1} \sum_i \hat{\eta}(X_i)$ of (3), plus the empirical average of (43) with the nuisances estimated,

$$\hat{p}_t^{\text{pi}}(y) + \frac{1}{n} \sum_{i=1}^n \left[\frac{K_{h_T}(T_i - t)}{\hat{e}(t | X_i)} (K_{h_Y}(Y_i - y) - \hat{\eta}(X_i)) + \hat{\eta}(X_i) - \hat{p}_t^{\text{pi}}(y) \right]. \quad (44)$$

The last two terms inside the sum average, respectively, to $\hat{p}_t^{\text{pi}}(y)$ itself and to $-\hat{p}_t^{\text{pi}}(y)$, and these cancel against the leading plug-in term: the plug-in does not disappear from the construction, it cancels out of the *display*. What remains of (44) is exactly the classical estimator (9), with no plug-in term visible in it, which is why reading (9) on its own can look unrelated to the one-step recipe of (7) that produced it. Under the smoothness and undersmoothing conditions of Appendix B.1, a second-order Taylor expansion of the smoothed functional around (t, y) gives $\mathbb{E}[\text{IF}_h(W; p_t(y), F)] = \mathcal{O}(h_T^2 + h_Y^2)$, so the substitution introduces a deterministic bias of that order, which the undersmoothing conditions render negligible on the $\sqrt{n}$ scale.

We now derive the bias claim in detail, since it is what licenses the substitution. Because $\mathbb{E}_F[\eta(X) - p_t(y)] = 0$ exactly, by Definition 1, only the first term of (43) contributes to $\mathbb{E}_F[\text{IF}_h(W; p_t(y), F)]$, and iterating expectations over X , then $T \mid X$, then $Y \mid T, X$ isolates

$$\mathbb{E}_F[\text{IF}_h(W; p_t(y), F)] = \mathbb{E}_X \left[\frac{1}{e(t \mid X)} \mathbb{E}_{T \mid X} \left[K_{h_T}(T - t) (\mathbb{E}_{Y \mid T, X} [K_{h_Y}(Y - y)] - \eta(X)) \right] \right]. \quad (45)$$

Fix $X = x$ and $T = \tau$. Substituting $v = (y' - y)/h_Y$ in the definition of the kernel average and Taylor-expanding the conditional density in its second argument around y , using that K_Y integrates to one, is symmetric (so $\int v K_Y(v) dv = 0$), and has finite second moment $\kappa_{2,Y} := \int v^2 K_Y(v) dv$,

$$\mathbb{E}_{Y \mid T=\tau, X=x} [K_{h_Y}(Y - y)] = \int K_Y(v) f_{Y \mid T=\tau, X=x}(y + h_Y v) dv = f_{Y \mid T=\tau, X=x}(y) + \frac{h_Y^2}{2} \kappa_{2,Y} \partial_y^2 f_{Y \mid T=\tau, X=x}(y) + o(h_Y^2),$$

uniformly over τ in a neighbourhood of t by the smoothness of Appendix B.1. Subtracting $\eta(x) = f_{Y \mid T=t, X=x}(y)$ and writing $g(\tau) := f_{Y \mid T=\tau, X=x}(y)$ for brevity,

$$\mathbb{E}_{Y \mid T=\tau, X=x} [K_{h_Y}(Y - y)] - \eta(x) = (g(\tau) - g(t)) + \frac{h_Y^2}{2} \kappa_{2,Y} g''(\tau) + o(h_Y^2).$$

Substitute $\tau = t + h_T \sigma$ in the outer, T -side average. By the same Taylor argument, now in the first argument of g , $g(t + h_T \sigma) - g(t) = h_T \sigma g'(t) + \mathcal{O}(h_T^2)$, and $f_{T \mid X=x}(t + h_T \sigma) = e(t \mid x) + \mathcal{O}(h_T)$. The inner average over $T \mid X = x$ becomes, after the substitution $\sigma = (\tau - t)/h_T$,

$$\mathbb{E}_{T \mid X=x} [K_{h_T}(T - t)(g(T) - \eta(x))] = \int K_T(\sigma) f_{T \mid X=x}(t + h_T \sigma) \left[h_T \sigma g'(t) + \frac{h_Y^2}{2} \kappa_{2,Y} g''(t) + \mathcal{O}(h_T^2 + h_Y^2) \right] d\sigma.$$

Expanding the product and integrating term by term: the leading piece $e(t \mid x) h_T g'(t) \int \sigma K_T(\sigma) d\sigma$ vanishes because K_T is symmetric, which is exactly why a second-order kernel is required; the cross term between the $\mathcal{O}(h_T)$ correction to $f_{T \mid X=x}$ and the $h_T \sigma g'(t)$ piece is $\mathcal{O}(h_T^2)$ by the same vanishing first moment; and the remaining terms collect into

$$\mathbb{E}_{T \mid X=x} [K_{h_T}(T - t)(g(T) - \eta(x))] = e(t \mid x) \frac{h_Y^2}{2} \kappa_{2,Y} \partial_y^2 \eta(x) + \mathcal{O}(h_T^2 + h_Y^2),$$

using $g''(t) = \partial_y^2 \eta(x)$. Substituting into (45), the factor $e(t \mid x)$ cancels the $1/e(t \mid X)$ weight, and averaging the remainder over X gives

$$\mathbb{E}_F[\text{IF}_h(W; p_t(y), F)] = \frac{h_Y^2}{2} \kappa_{2,Y} \mathbb{E}_X [\partial_y^2 \eta(X)] + \mathcal{O}(h_T^2 + h_Y^2) = \mathcal{O}(h_T^2 + h_Y^2),$$

which is the claimed bias. The symmetric, second-order kernel is used twice: to remove the first-order term in y (via $\int v K_Y = 0$) and, independently, to remove the first-order term in t (via $\int \sigma K_T = 0$); a first-order kernel would leave a bias of order h_T or h_Y , too slow for the undersmoothing conditions of Appendix B.1 to control.

Two special cases recover known influence functions, and reproducing them checks the derivation. If T takes finitely many values, the point mass $\delta_{t_w}(t)$ is the indicator $\mathbf{1}(t_w = t)$, no smoothing is needed in the treatment argument, and $e(t \mid x)$ is the ordinary propensity score; (43) then reduces to the influence function of Kennedy et al. (2023). If the target is $\mathbb{E}[Y(t)]$, the outcome is integrated rather than evaluated: the derivation carries an extra factor y and an extra integral in y , the point mass $\delta_{y_w}(y)$ is replaced by y_w , and (43) reduces to the influence function of Kennedy et al. (2017).

B.4 Second-order remainder

We prove Proposition 2. Let $\widehat{e}$ and $\widehat{f}_{Y|T,X}$ be nuisance estimates fitted on an independent fold, write $\eta(x) := f_{Y|T=t,X=x}(y)$ and $\widehat{\eta}(x) := \widehat{f}_{Y|T=t,X=x}(y)$, and condition throughout on the fitting fold, so that $\widehat{e}, \widehat{\eta}$ are fixed and the outer expectation $\mathbb{E}$ below is over the evaluation fold alone, at the true law F . Define the oracle functional, for any candidate pair (g, e) ,

$$\theta(g, e) := \mathbb{E} \left[\frac{K_{h_T}(T-t)}{e(t|X)} (K_{h_Y}(Y-y) - g(X)) + g(X) \right],$$

so that the classical estimator satisfies $\mathbb{E}[\widehat{p}_t^c(y) | \text{fitting fold}] = \theta(\widehat{\eta}, \widehat{e})$, while $\theta(\eta, e) = \mathbb{E}_F[\text{IF}_h(W; p_t(y), F)] + p_t(y)$ is the smoothed oracle of Appendix B.3, equal to $p_t(y) + \mathcal{O}(h_T^2 + h_Y^2)$. The remainder we must bound is $\text{Rem}_2 := \theta(\widehat{\eta}, \widehat{e}) - \theta(\eta, e) - \mathcal{O}(h_T^2 + h_Y^2)$, and we compute the difference of the two oracles exactly.

Subtracting the two functionals and expanding the definition of θ ,

$$\theta(\widehat{\eta}, \widehat{e}) - \theta(\eta, e) = \mathbb{E} \left[K_{h_T}(T-t) \left\{ \frac{K_{h_Y}(Y-y) - \widehat{\eta}(X)}{\widehat{e}(t|X)} - \frac{K_{h_Y}(Y-y) - \eta(X)}{e(t|X)} \right\} \right] + \mathbb{E}[\widehat{\eta}(X) - \eta(X)]. \quad (46)$$

Write $a := K_{h_Y}(Y-y)$, $e := e(t|X)$, $\widehat{e} := \widehat{e}(t|X)$ for brevity. The identity $(a - \widehat{\eta})/\widehat{e} - (a - \eta)/e = (a - \eta)(1/\widehat{e} - 1/e) + (\eta - \widehat{\eta})/\widehat{e}$, obtained by adding and subtracting $\eta/\widehat{e}$, splits the bracket in (46) into two channels, so that (46) becomes $T_1 + T_2 + T_3$ with

$$T_1 := \mathbb{E} \left[K_{h_T}(T-t) (K_{h_Y}(Y-y) - \eta(X)) \left(\frac{1}{\widehat{e}(t|X)} - \frac{1}{e(t|X)} \right) \right], \quad (47)$$

$$T_2 := \mathbb{E} \left[K_{h_T}(T-t) (\eta(X) - \widehat{\eta}(X)) / \widehat{e}(t|X) \right], \quad (48)$$

$$T_3 := \mathbb{E}[\widehat{\eta}(X) - \eta(X)]. \quad (49)$$

Each channel is evaluated by conditioning on X , then on T , exactly as in Appendix B.3. For $T_2 + T_3$, condition on X and take the inner expectation over $T | X$ first: $\mathbb{E}_{T|X}[K_{h_T}(T-t)] = e(t|X) + \mathcal{O}(h_T^2)$ by the symmetric-kernel argument of Appendix B.3 applied to e in place of η , so

$$T_2 + T_3 = \mathbb{E} \left[(\eta(X) - \widehat{\eta}(X)) \left(\frac{e(t|X)}{\widehat{e}(t|X)} - 1 \right) \right] + \mathcal{O}(h_T^2) = \mathbb{E} \left[(\widehat{\eta}(X) - \eta(X)) \frac{e(t|X) - \widehat{e}(t|X)}{\widehat{e}(t|X)} \right] + \mathcal{O}(h_T^2).$$

The $\mathcal{O}(h_T^2)$ term is of smaller order than the product of nuisance errors under the rate condition below, and is absorbed into the bias already controlled by Appendix B.3. For T_1 , the same conditioning shows the bracket $K_{h_Y}(Y-y) - \eta(X)$ has conditional mean, given (T, X) , equal to $f_{Y|T,X}(y) - \eta(X) + \mathcal{O}(h_Y^2)$, which is itself $\mathcal{O}(|T-t|)$ inside the effective support of $K_{h_T}(T-t)$ by smoothness of $f_{Y|T,X}(y)$ in t (Appendix B.1); hence $T_1 = \mathcal{O}(h_T + h_Y^2)$ times a nuisance-error average, one order smaller in h than $T_2 + T_3$, and negligible relative to it. Collecting the dominant channel gives

$$\text{Rem}_2 = \mathbb{E} \left[\left(\frac{K_{h_T}(T-t)}{\widehat{e}(t|X)} - \frac{K_{h_T}(T-t)}{e(t|X)} \right) (\eta(X) - \widehat{\eta}(X)) \right] + \mathcal{O}(h_T^2), \quad (50)$$

recovering the cross term claimed above, up to the kernel-bias term already accounted for. This is a product of the two nuisance errors: the first factor is the error in the inverse propensity weight and the second is the error in the conditional density. Applying the Cauchy–Schwarz inequality

and bounding $\widehat{e}^{-1}$ and e^{-1} using the lower bound $e_{\min}$ of Assumption 1(iii) and the boundedness of K_{h_T} ,

$$|\text{Rem}_2| \lesssim \|\widehat{e} - e\|_{L_2} \|\widehat{\eta} - \eta\|_{L_2} + \mathcal{O}(h_T^2).$$

The remainder is therefore controlled by the product of the two L_2 errors, plus a kernel-bias term already of order $\mathcal{O}(h_T^2)$ under Appendix B.1. It is $o(n^{-1/2})$ whenever each nuisance error is $o(n^{-1/4})$, and one error may be larger if the other is correspondingly smaller, so that the product still vanishes faster than $n^{-1/2}$; this is exactly the double robustness asserted in Proposition 2. $\square$

C The contrastive correction

This appendix proves the results of Section 4.2. We first prove the reconstruction identity, then differentiate the contrastive factorization to obtain Proposition 4.

C.1 Proof of Proposition 3

From Definition 4, the odds of the contrast function are the ratio of its two mixture components,

$$\frac{D_\pi}{1 - D_\pi} = \frac{\pi f_{T,X,Y}(t, x, y)}{(1 - \pi) f_{T,X}(t, x) f_Y(y)} = \frac{\pi}{1 - \pi} \frac{f_{Y|T=t, X=x}(y)}{f_Y(y)},$$

where the second equality uses the factorization $f_{T,X,Y}(t, x, y) = f_{T,X}(t, x) f_{Y|T=t, X=x}(y)$ to cancel $f_{T,X}(t, x)$. Multiplying both sides by $(1 - \pi)/\pi$ isolates $\frac{1-\pi}{\pi} D_\pi / (1 - D_\pi) = f_{Y|T=t, X=x}(y) / f_Y(y)$, and by the definition of R in (13) the left-hand side is $R(t, x, y)$. Rearranging gives both $R = \frac{1-\pi}{\pi} D_\pi / (1 - D_\pi)$ and $f_{Y|T=t, X=x}(y) = f_Y(y) R$, which are the two identities of (15). $\square$

C.2 Proof of Proposition 4

We differentiate the contrastive factorization (17), $p_t(y) = f_Y(y) m(t, y)$. By the product rule, Lemma 3,

$$\text{IF}(w; p_t(y), F) = f_Y(y) \text{IF}(w; m(t, y), F) + m(t, y) \text{IF}(w; f_Y(y), F). \quad (51)$$

We compute the two influence functions on the right in turn.

For the marginal factor, $f_Y(y)$ depends on F only through the marginal law of the outcome, evaluated at the fixed point y . This is the univariate special case of the computation in Appendix B.2: differentiating a marginal density at a fixed point along the contamination path returns the difference between a point mass and the density,

$$\text{IF}(w; f_Y(y), F) = \delta_{y_w}(y) - f_Y(y), \quad (52)$$

which is regularized by $\delta_{y_w}(y) \rightsquigarrow K_{h_Y}(y_w - y)$ as in Appendix B.3.

For the covariate-averaged factor $m(t, y) = \mathbb{E}_X[R(t, X, y)]$, the integrand $R(t, X, y; F)$ depends on F both through the law of X and through its own functional form, since the density ratio is itself a functional of F . Lemma 7 is written for exactly this situation, and gives

$$\text{IF}(w; m(t, y), F) = R(t, x_w, y) - m(t, y) + \mathbb{E}_X[\text{IF}(w; R(t, X, y), F)]. \quad (53)$$

The third term is present precisely because the density ratio is fitted rather than known; it is the term that will carry the influence function of the classifier. Substituting the odds rule (38) of Lemma 6 for $\text{IF}(w; R, F)$ inside that expectation turns it into $\frac{1-\pi}{\pi} \mathbb{E}_X[\text{IF}(w; D_\pi, F) / (1 - D_\pi)^2]$.

Finally, substitute the marginal influence function (52) and the covariate influence function (53), with the odds rule applied, into the product (51):

$$\text{IF}(w; p_t(y), F) = f_Y(y) \left[R(t, x_w, y) - m(t, y) + \frac{1 - \pi}{\pi} \mathbb{E}_X \left[\frac{\text{IF}(w; D_\pi, F)}{(1 - D_\pi)^2} \right] \right] + m(t, y) (\delta_{y_w}(y) - f_Y(y)).$$

Grouping the first three terms, all multiplied by $f_Y(y)$, inside a single brace and leaving the last term, carried by $m(t, y) \text{IF}(w; f_Y(y), F)$, on its own is exactly (18), the claim of Proposition 4. $\square$

D The differentiable discriminant

This appendix proves the results of Section 4.5 and the specialization of Section 5. Appendix D.1 records the gradients, the Hessian and the convexity, and proves Proposition 6. Appendix D.2 differentiates the two component laws. Appendix D.3 assembles these into Theorem 1. Appendices D.4 and D.5 prove the closed form (Proposition 5) and the zero-mean property (Lemma 1), and Appendix D.6 proves Corollary 1. Throughout, $\beta = \beta_\lambda(F)$, $D := D_\beta$, and all expectations are taken at F .

One convention keeps the displays readable. We write $\phi := \phi(t, x, y)$ for the logit Jacobian J_β throughout. Only Appendix D.1 uses that it is a *fixed* map, through $\nabla_\beta^2 F_\beta \equiv 0$; that subsection is therefore the one place where the argument is specific to the affine logit of Section 5, and it is exactly where Proposition 6 is proved. No other derivation below ever differentiates ϕ , so every one of them holds verbatim with ϕ read as J_β , which is what makes the results of Section 4.5 general and Section 6 short.

D.1 Gradients, Hessian and convexity

Differentiating the logistic loss in its first argument,

$$\frac{\partial \ell}{\partial D}(D, z) = -\frac{z}{D} + \frac{1 - z}{1 - D} = \frac{D - z}{D(1 - D)}. \quad (54)$$

Differentiating the model in β , with $\sigma' = \sigma(1 - \sigma)$,

$$\nabla_\beta D_\beta = D(1 - D) \phi. \quad (55)$$

Composing the two by the chain rule, the factor $D(1 - D)$ cancels and the loss gradient simplifies to

$$\nabla_\beta \ell(D_\beta, z) = (D - z) \phi, \quad \text{so} \quad u_\beta = (D - 1) \phi, \quad v_\beta = D \phi, \quad (56)$$

where u_β and v_β are the gradients at labels $z = 1$ and $z = 0$. Differentiating once more and using (55), both labels give the same second derivative $\nabla_\beta u_\beta = \nabla_\beta v_\beta = D(1 - D) \phi \phi^\top$, so the Hessian of the penalized risk is

$$H_\lambda = \mathbb{E}_{P_\pi} [D(1 - D) \phi \phi^\top] + 2\lambda I_k. \quad (57)$$

For any $s \in \mathbb{R}^k$,

$$s^\top H_\lambda s = \mathbb{E}_{P_\pi} [D(1 - D)(s^\top \phi)^2] + 2\lambda \|s\|^2 \geq 2\lambda \|s\|^2,$$

since $D(1 - D) > 0$ makes the first term nonnegative. Hence H_λ is positive definite with every eigenvalue at least 2λ , the criterion (22) is strictly convex, and $\beta_\lambda(F)$ exists, is unique and is interior. When $\lambda = 0$ the same computation gives positive definiteness provided $\mathbb{E}_{P_\pi} [(s^\top \phi)^2] > 0$ for every $s \neq 0$, i.e. provided ϕ is not collinear on the support. The first-order condition $\nabla_\beta \mathcal{R}_\lambda = 0$ reads, using (56),

$$G_\lambda(\beta; F) := \pi \mathbb{E}_{P_1} [u_\beta] + (1 - \pi) \mathbb{E}_{P_0} [v_\beta] + 2\lambda \beta = 0. \quad (58)$$

D.2 Perturbation of the two component laws

We differentiate each component of the mixture P_π along the contamination path, treating β as fixed; the dependence of β on F is restored in Appendix D.3.

The positive law is the observed law, $P_1(F_\varepsilon) = F_\varepsilon$, so $\mathbb{E}_{F_\varepsilon}[u] = (1 - \varepsilon)\mathbb{E}_F[u] + \varepsilon u(w)$ is affine in ε , and its derivative is

$$\partial_\varepsilon \mathbb{E}_{P_1(F_\varepsilon)}[u] \Big|_0 = u(w) - \mathbb{E}_{P_1}[u]. \quad (59)$$

The negative law is the product $P_0(F_\varepsilon) = F_{T,X,\varepsilon} \otimes F_{Y,\varepsilon}$, whose two factors are the contaminated marginals of F_ε ,

$$F_{T,X,\varepsilon} = (1 - \varepsilon)F_{T,X} + \varepsilon \delta_{(t_w, x_w)}, \quad F_{Y,\varepsilon} = (1 - \varepsilon)F_Y + \varepsilon \delta_{y_w}.$$

Expanding the double integral $\mathbb{E}_{P_0(F_\varepsilon)}[v]$ into the four terms according to which factor carries the contamination,

$$\begin{aligned} \mathbb{E}_{P_0(F_\varepsilon)}[v] &= (1 - \varepsilon)^2 \mathbb{E}_{P_0}[v] + (1 - \varepsilon)\varepsilon \mathbb{E}_{F_{T,X}}[v(T, X, y_w)] \\ &\quad + \varepsilon(1 - \varepsilon) \mathbb{E}_{F_Y}[v(t_w, x_w, \tilde{Y})] + \varepsilon^2 v(t_w, x_w, y_w). \end{aligned} \quad (60)$$

Differentiate at $\varepsilon = 0$. The quadratic term $\varepsilon^2 v(t_w, x_w, y_w)$ has zero derivative; the factor $(1 - \varepsilon)^2$ contributes $-2 \mathbb{E}_{P_0}[v]$; and each of the two cross terms, being $\varepsilon(1 - \varepsilon)$ times a constant, contributes that constant. Hence

$$\partial_\varepsilon \mathbb{E}_{P_0(F_\varepsilon)}[v] \Big|_0 = \mathbb{E}_{F_Y}[v(t_w, x_w, \tilde{Y})] + \mathbb{E}_{F_{T,X}}[v(T, X, y_w)] - 2 \mathbb{E}_{P_0}[v]. \quad (61)$$

The coefficient -2 on the centring term is what splits the contrastive derivative into two channels, one for each factor of the product measure. Equivalently, applying the product rule of Lemma 3 to the bilinear map $(\mu, \nu) \mapsto \iint v \, d\mu \, d\nu$ produces the first bracket by differentiating $F_{T,X,\varepsilon}$ at fixed F_Y and the second by differentiating $F_{Y,\varepsilon}$ at fixed $F_{T,X}$. The mixture ratio π is a design constant and contributes no derivative.

D.3 Proof of Theorem 1

By definition the perturbed minimizer $\beta_\lambda(F_\varepsilon)$ solves the first-order condition $G_\lambda(\beta_\lambda(F_\varepsilon); F_\varepsilon) = 0$ identically in ε . The left-hand side depends on ε in two ways, through the parameter $\beta_\lambda(F_\varepsilon)$ and through the law F_ε . Differentiating the identity at $\varepsilon = 0$ by the implicit function theorem,

$$\underbrace{\nabla_\beta G_\lambda}_{= H_\lambda} \underbrace{\partial_\varepsilon \beta_\lambda(F_\varepsilon) \Big|_0}_{= \text{IF}(w; \beta_\lambda, F)} + \underbrace{\partial_\varepsilon G_\lambda(\beta; F_\varepsilon) \Big|_0}_{=: S(w; \beta)} = 0. \quad (62)$$

The first factor is $\nabla_\beta G_\lambda = H_\lambda$ by (57), and it is invertible by Assumption 2, which Appendix D.1 establishes outright in the affine case; solving gives $\text{IF}(w; \beta_\lambda, F) = -H_\lambda^{-1} S(w; \beta)$, which is the first equation of (25). It remains to identify $S = \partial_\varepsilon G_\lambda \Big|_0$. Since $G_\lambda(\beta; F_\varepsilon) = \pi \mathbb{E}_{P_1(F_\varepsilon)}[u_\beta] + (1 - \pi) \mathbb{E}_{P_0(F_\varepsilon)}[v_\beta] + 2\lambda\beta$ and the penalty $2\lambda\beta$ does not depend on F , applying (59) with $u = u_\beta$ to the first expectation and (61) with $v = v_\beta$ to the second gives the three summands

$$S = \pi(u_\beta(w) - \mathbb{E}_{P_1}[u_\beta]) + (1 - \pi)(\mathbb{E}_{F_Y}[v_\beta(t_w, x_w, \tilde{Y})] - \mathbb{E}_{P_0}[v_\beta]) + (1 - \pi)(\mathbb{E}_{F_{T,X}}[v_\beta(T, X, y_w)] - \mathbb{E}_{P_0}[v_\beta]),$$

which are exactly S_1 , S_0^{TX} and S_0^Y of (26). This establishes both equations of (25) and hence Theorem 1. $\square$

D.4 Proof of Proposition 5

We evaluate the three channels for the logistic model. By (56), $v_\beta = D\phi$, so every centring term $\mathbb{E}_{P_0}[v_\beta]$ equals $\mathbb{E}_{P_0}[D\phi] = b_0$, and the two contaminated contrastive terms are $b_{TX}(t_w, x_w)$ and $b_Y(y_w)$ of (29). Summing the three channels of (26) and grouping the two centring terms with the observed centring term,

$$S(w; \beta) = \pi u_\beta(w) + (1 - \pi)(b_{TX}(t_w, x_w) + b_Y(y_w)) - (\pi \mathbb{E}_{P_1}[u_\beta] + 2(1 - \pi)b_0). \quad (63)$$

The bracketed quantity contains the unknown expectation $\mathbb{E}_{P_1}[u_\beta]$, which the first-order condition (58) pins down: rearranging $\pi \mathbb{E}_{P_1}[u_\beta] + (1 - \pi)b_0 + 2\lambda\beta_\lambda = 0$ gives $\pi \mathbb{E}_{P_1}[u_\beta] = -2\lambda\beta_\lambda - (1 - \pi)b_0$. Substituting this into the bracket,

$$- (\pi \mathbb{E}_{P_1}[u_\beta] + 2(1 - \pi)b_0) = 2\lambda\beta_\lambda + (1 - \pi)b_0 - 2(1 - \pi)b_0 = 2\lambda\beta_\lambda - (1 - \pi)b_0. \quad (64)$$

The uncanceled residual $-(1 - \pi)b_0$ appears because the first-order condition supplies a single contrastive centring term whereas the channels of Theorem 1 produce two. Combining (63) and (64) and writing $u_\beta = (D - 1)\phi$ from (56) gives

$$S(w; \beta) = \pi(D_\beta(w) - 1)\phi(w) + (1 - \pi)(b_{TX}(t_w, x_w) + b_Y(y_w) - b_0) + 2\lambda\beta_\lambda,$$

which is exactly (30), the claim of Proposition 5. $\square$

Two limiting cases are worth recording, as they connect the formula to familiar ones. Setting $\lambda = 0$ removes the penalty term $2\lambda\beta_\lambda$. Setting in addition the two contrastive channels to zero, the situation of a negative class drawn from a distribution fixed by the analyst, so that b_{TX} , b_Y and b_0 carry no dependence on F , leaves $S(w; \beta) = \pi(D_\beta(w) - 1)\phi(w)$, and (25) becomes $-H_0^{-1}(D_\beta(w) - z_w)\phi(w)$, the classical influence function of logistic regression.

D.5 Proof of Lemma 1

We take the expectation of the closed form (30) over $W \sim F$. The key structural fact is that when $W = (T, X, Y)$ is drawn from F , its first two components jointly follow $F_{T,X}$ and its last component follows F_Y , the same marginals that define P_0 .

The first term is an expectation under $P_1 = F_{T,X,Y}$, so it equals $\pi \mathbb{E}_{P_1}[(D - 1)\phi] = \pi \mathbb{E}_{P_1}[u_\beta]$ by (56).

The vector $b_{TX}(T, X) = \mathbb{E}_{\tilde{Y} \sim F_Y}[v_\beta(T, X, \tilde{Y})]$ is a function of (T, X) alone. Averaging it over $(T, X) \sim F_{T,X}$ and using Fubini together with the product structure of P_0 ,

$$\mathbb{E}[b_{TX}(T, X)] = \mathbb{E}_{(T,X) \sim F_{T,X}} \mathbb{E}_{\tilde{Y} \sim F_Y}[v_\beta(T, X, \tilde{Y})] = \mathbb{E}_{P_0}[v_\beta] = b_0.$$

Symmetrically, $b_Y(Y) = \mathbb{E}_{(T,X) \sim F_{T,X}}[v_\beta(T, X, Y)]$ is a function of Y alone and, averaged over $Y \sim F_Y$, also equals b_0 .

The vectors b_0 and $2\lambda\beta_\lambda$ are constants and average to themselves. Summing the four contributions,

$$\mathbb{E}[S(W; \beta)] = \pi \mathbb{E}_{P_1}[u_\beta] + (1 - \pi)(b_0 + b_0 - b_0) + 2\lambda\beta_\lambda = \pi \mathbb{E}_{P_1}[u_\beta] + (1 - \pi)b_0 + 2\lambda\beta_\lambda,$$

which is exactly $G_\lambda(\beta_\lambda; F)$ and therefore vanishes by the first-order condition (58). This proves $\mathbb{E}_{W \sim F}[S(W; \beta_\lambda)] = 0$. Since H_λ does not depend on w , taking the expectation of $\text{IF}(w; \beta_\lambda, F) = -H_\lambda^{-1}S(w; \beta_\lambda)$ gives $\mathbb{E}_{W \sim F}[\text{IF}(W; \beta_\lambda, F)] = -H_\lambda^{-1}\mathbb{E}[S] = 0$, which is the second statement of Lemma 1. $\square$

The computation also shows what would go wrong without the residual: had the term $-(1-\pi)b_0$ been omitted from (30), the contrastive contributions would have summed to $2b_0$ instead of b_0 , and the total mean would have been $(1-\pi)b_0 \neq 0$, which is impossible for an influence function.

D.6 Proof of Corollary 1

We collapse the correction term to an inner product. By (31) together with the model gradient (55),

$$\text{IF}(w; D_\pi(t, x, y), F) = \nabla_\beta D_\beta(t, x, y)^\top \text{IF}(w; \beta_\lambda, F) = D(1-D) \phi(t, x, y)^\top \text{IF}(w; \beta_\lambda, F).$$

Divide by $(1-D)^2$, multiply by $(1-\pi)/\pi$, and take the expectation over the covariates. Because $\text{IF}(w; \beta_\lambda, F)$ does not depend on X , it factors out of the expectation, and the scalar weight simplifies to the density ratio,

$$\frac{1-\pi}{\pi} \frac{D(1-D)}{(1-D)^2} = \frac{1-\pi}{\pi} \frac{D}{1-D} = R,$$

by the definition $R_\beta := \frac{1-\pi}{\pi} D_\beta / (1-D_\beta)$. Hence

$$\frac{1-\pi}{\pi} \mathbb{E}_X \left[\frac{\text{IF}(w; D_\pi(t, X, y), F)}{(1-D_\pi(t, X, y))^2} \right] = \mathbb{E}_X [R(t, X, y) \phi(t, X, y)]^\top \text{IF}(w; \beta_\lambda, F) = a_\lambda(t, y)^\top \text{IF}(w; \beta_\lambda, F),$$

which is (32), the claim of Corollary 1. $\square$

Substituting $\text{IF}(w; \beta_\lambda, F) = -H_\lambda^{-1} S(w; \beta)$ and using the symmetry of H_λ , with $\xi(t, y) := H_\lambda^{-1} a_\lambda(t, y)$,

$$a_\lambda(t, y)^\top \text{IF}(w; \beta_\lambda, F) = - (H_\lambda^{-1} a_\lambda(t, y))^\top S(w; \beta) = -\xi(t, y)^\top S(w; \beta),$$

which is the form used in the estimator (33) and requires one linear solve per evaluation point rather than one per observation.

E Asymptotic analysis

This appendix proves the normalization constraint of Section 8.1; the asymptotic analysis proper is left for later. **[TO COMPLETE: À voir.]**

E.1 Proof of Proposition 8

Integrating the joint density over (t, x) recovers the marginal of the outcome, and factoring through the conditional density,

$$f_Y(y) = \iint f_{T,X,Y}(t, x, y) dt dx = \iint f_{T,X}(t, x) f_{Y|T=t, X=x}(y) dt dx.$$

Substituting the reconstruction identity (15), $f_{Y|T=t, X=x}(y) = f_Y(y) R(t, x, y)$, and taking the constant $f_Y(y)$ outside the integral,

$$f_Y(y) = f_Y(y) \iint f_{T,X}(t, x) R(t, x, y) dt dx = f_Y(y) \mathbb{E}_{(T,X)} [R(T, X, y)].$$

Since $f_Y(y) > 0$, dividing both sides by it gives $\mathbb{E}_{(T,X)} [R(T, X, y)] = 1$, which is (37), the claim of Proposition 8. $\square$

Written in terms of the contrast function, the identity reads $\mathbb{E}_{(T,X)} [D_\pi(T, X, y) / (1-D_\pi(T, X, y))] = \pi / (1-\pi)$ for every such y . It holds for each y separately and therefore constitutes a continuum of restrictions linking the contrast function to the marginal density of the outcome, which is the sense in which the two factors of the correction are not variationally independent.

F Data-generating processes

This appendix gives the full specification of every data-generating process used in Section 7, referenced there only by name.

F.1 The AXY dataset

AXY is a binary-treatment, high-dimensional-covariate design, adapted from the toy structural model of Melnychuk et al. (2023) (their Figure 1 and Appendix H.1): its two treatment arms and their quadratic-versus-linear asymmetry are theirs, generalized here from a scalar covariate to a projection of a cov_dim -dimensional one, so that the covariate dimension can be swept while the treatment stays binary:

$$\begin{aligned} X &\sim 0.5\mathcal{N}(0, I_{\text{cov_dim}}) + 0.5\mathcal{N}(\text{cov_shift} \cdot w, I_{\text{cov_dim}}), & w &:= \text{cov_dim}^{-1/2}\mathbf{1}, \\ e(1 \mid x) &= \frac{\mathcal{N}(x; 0, I)}{\mathcal{N}(x; 0, I) + \mathcal{N}(x; \text{cov_shift} \cdot w, I)}, & T \mid X = x &\sim \text{Bernoulli}(e(1 \mid x)), \\ Y &= U_Y + \begin{cases} Z^2 - 1.82Z + 2 & \text{if } T = 1 \\ 2.18Z + 1.5 & \text{if } T = 0 \end{cases}, & Z &:= w^\top X, \quad U_Y \sim \mathcal{N}(0, 1), \end{aligned} \tag{65}$$

with $e(\cdot \mid x)$ available in closed form by construction, the one property this design is used for in Section 7.3: it lets the classical estimator (9) be computed directly, without a further nuisance-estimation step, and checked against the IF-corrected one (33) up to sampling error. cov_shift controls the overlap between the two covariate sub-populations, exactly as b does in Melnychuk et al. (2023)’s original design: small values give near-complete overlap, large values push $e(1 \mid x)$ towards 0 or 1.

w is the unit vector along the shift between the two mixture components, so $Z = w^\top X$ is a one-dimensional sufficient statistic for the treatment mechanism, and its marginal law is exactly $0.5\mathcal{N}(0, 1) + 0.5\mathcal{N}(\text{cov_shift}, 1)$: *independent of cov_dim*, since $w^\top w = 1$ by construction. Consequently $p_t(y)$ (Figure 2) does not change shape as cov_dim grows: growing the covariate dimension changes nothing about the *target*, only about how hard e and the conditional density are to estimate from a finite sample, which is precisely why this design isolates the effect of confounder dimensionality (Section 8.2) from the effect of a harder target.

F.2 The well-specified linear-Gaussian model

Every design above and below is deliberately misspecified for a low-degree polynomial feature map: AXY’s covariate is a Gaussian mixture, and each of the four models of Appendix F.3 adds a property (multimodality, heavy tails, heteroscedasticity, skewness) that a Gaussian log-density cannot produce. This leaves open a question none of them can answer: does the estimator behave correctly when nothing is fighting the feature map, only sampling noise? A confounder $Z \sim \mathcal{N}(\mu_Z, \Sigma_Z) \in \mathbb{R}^{\text{cov_dim}}$ and a linear-Gaussian outcome answer it,

$$Y \mid T = t, Z = z \sim \mathcal{N}(a^\top z + b^\top t, \sigma^2), \tag{66}$$

with T continuous and of any dimension in this design’s native form, matching the treatment this paper targets; the runs of Section 7.3 constrain it to a binary T only because the baselines of Appendix J it is compared against there require a discrete treatment, exactly the reason AXY is

Shared outcome mechanism of the toy SCM and AXY (Appendices H.1, H.3)

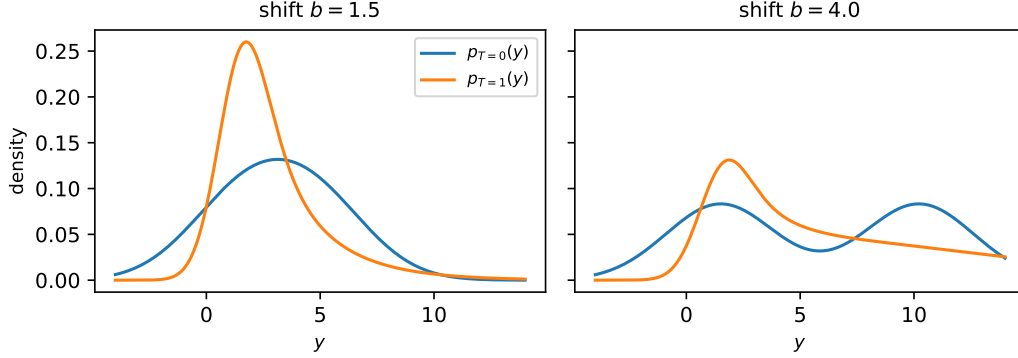


Figure 2: AXY’s outcome mechanism (65): potential-outcome densities $p_{T=0}(y)$ (linear arm) and $p_{T=1}(y)$ (quadratic arm) at two values of cov_shift , both identical across every cov_dim as shown in the text above. At the larger shift, the linear arm inherits the bimodality of the underlying covariate mixture, while the quadratic arm folds both mixture components onto overlapping regions near its vertex and stays unimodal, the same mixture confounder producing qualitatively different outcome shapes depending only on the (non)linearity of the outcome mechanism.

binary, not a limitation of the design itself. Since $\text{do}(T = t)$ severs $Z \rightarrow T$, Z reverts to its own marginal exactly as in Appendix F.3, giving

$$Y \mid \text{do}(T = t) \sim \mathcal{N}(a^\top \mu_Z + b^\top t, a^\top \Sigma_Z a + \sigma^2), \quad (67)$$

exactly, with no quadrature or Monte Carlo step anywhere in the computation, a stronger sense of closed form than any other design here, since even the elementary skew-normal case needs the adjusted-parameter substitution of (76) and AXY’s quadratic arm needs a numerical integral over its mixture confounder.

This is also the only design in which the contrast function’s log-odds is an *exact* low-degree polynomial. Both $f(y \mid t, z)$ and, because Z and T are jointly Gaussian under the sampling law, the marginal $f_Y(y)$ are Gaussian densities, so $\log f(y \mid t, z) - \log f_Y(y)$ is exactly a quadratic form in (t, z, y) : every term is degree two or less, with no exponential, mixture, or Gamma-mixture structure left over the way there is in every other design above. A degree-two polynomial feature map (Table 3’s feature-map-degree grid) therefore represents it without approximation error, so that $D_{\beta_\lambda} = D_\pi$ exactly rather than only in the limit. This is precisely the condition under which nothing is left to control on the approximation side: this design isolates the estimation-only regime, the one AXY and the four models of Appendix F.3 cannot exercise on their own, since a low-degree polynomial is misspecified for all of them.

F.3 The four continuous, multidimensional-treatment models

All four models below share the same confounding structure: (X, T) jointly Gaussian,

$$\begin{pmatrix} X \\ T \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_T \end{pmatrix}, \begin{pmatrix} \Sigma_X & \Sigma_{XT} \\ \Sigma_{XT}^\top & \Sigma_T \end{pmatrix}\right), \quad X \in \mathbb{R}^p, T \in \mathbb{R}^d, \quad (68)$$

with $p = \dim(X)$ and $d = \dim(T)$ both free to vary, and Σ_{XT} controlling the strength of confounding. Since $\text{do}(T = t)$ severs $X \rightarrow T$, X reverts to its own marginal $\mathcal{N}(\mu_X, \Sigma_X)$ under the

intervention regardless of Σ_{XT} ; every closed form below uses only this marginal, never the conditional law of X given T . What varies across the four models is only the outcome law $Y \mid T, X$, since, as already stressed in Section 7.2, that is the only part of the model that can change the shape of $p_t(y)$.

Each of the four adds exactly one feature a jointly Gaussian (X, T, Y) cannot produce, and each feature is chosen because a purely Gaussian, linear-in- (t, x) specification of $Y \mid T, X$ — the assumption underlying a plug-in built on a conditional mean and a constant-variance noise model, represents it either not at all or only by contorting the mean and variance to compensate. Together the four probe whether the discriminant can recover a shape a mean-and-variance description would miss, before Section 7.3 asks the sharper question of *which* discriminant, and at what cost.

Multimodal. A two-component mixture whose *weight*, not only its component means, depends on t :

$$Y \mid T = t, X = x \sim w(t) \mathcal{N}(a_1^\top x + b_1^\top t, s_1^2) + (1 - w(t)) \mathcal{N}(a_2^\top x + b_2^\top t, s_2^2), \quad w(t) = \sigma(c_0 + c_1^\top t). \quad (69)$$

Because $w(t)$ does not depend on x , integrating x against its own marginal acts separately on each mixture component:

$$p_t(y) = w(t) \mathcal{N}(y; a_1^\top \mu_X + b_1^\top t, a_1^\top \Sigma_X a_1 + s_1^2) + (1 - w(t)) \mathcal{N}(y; a_2^\top \mu_X + b_2^\top t, a_2^\top \Sigma_X a_2 + s_2^2). \quad (70)$$

The difficulty parameters are the separation between the two components ($\|b_1 - b_2\|$, $\|a_1 - a_2\|$) and c_1 , how quickly the mixing weight moves with t : both jointly control how easy the resulting bimodality is to detect at a given sample size. A weight that moves with t makes $p_t(y)$ change *shape*, not only location or scale, as t varies, exactly the kind of dependence on t that a discriminant with too little capacity in the treatment coordinate cannot represent, which is what varying c_1 at fixed n is built to expose (Section 7.3).

Heavy-tailed. A linear-Gaussian mean with a standardized Student- t noise term in place of a Gaussian one:

$$Y \mid T = t, X = x = a^\top x + b^\top t + \sigma \varepsilon, \quad \varepsilon \sim \text{standardized Student-}t(\nu). \quad (71)$$

$p_t(y)$ is now the convolution of a Gaussian ($a^\top X$, $X \sim \mathcal{N}(\mu_X, \Sigma_X)$) with a scaled Student- t , which has no elementary closed form, but does have an exact semi-closed form through the Student- t 's Gamma precision-mixture representation ($\varepsilon \mid W \sim \mathcal{N}(0, c_\nu/W)$, $W \sim \text{Gamma}(\nu/2, \nu/2)$, mean 1, $c_\nu = (\nu - 2)/\nu$ for $\nu > 2$ the factor that standardizes ε to unit variance):

$$p_t(y) = \mathbb{E}_W \left[\mathcal{N}(y; a^\top \mu_X + b^\top t, a^\top \Sigma_X a + \sigma^2 c_\nu/W) \right], \quad W \sim \text{Gamma}(\nu/2, \nu/2). \quad (72)$$

The one remaining integral, over W alone, is evaluated by a deterministic quadrature at 400 equally spaced quantiles of the $\text{Gamma}(\nu/2, \nu/2)$ law, not by Monte Carlo: the result is exact to machine precision and identical across repeated evaluations, which is what makes it usable as ground truth. Smaller ν gives heavier tails; $\nu \rightarrow \infty$ recovers the plain Gaussian, linear model as a continuity check. A heavy tail is a property of $p_t(y)$ away from its mode, so this model stresses a different part of the estimator than the other three: the outcome marginal $\hat{f}_Y$ and bandwidth h_Y in the region where data are sparse, rather than the discriminant's fit near the bulk of the distribution.

Heteroscedastic. A variance that grows with t through a log link, guaranteeing positivity, with everything else linear-Gaussian:

$$Y \mid T = t, X = x \sim \mathcal{N}(a^\top x + b^\top t, \sigma_0^2 \exp(c^\top t)). \quad (73)$$

The variance does not depend on x , so integrating x against its marginal leaves the model Gaussian:

$$p_t(y) = \mathcal{N}(y; a^\top \mu_X + b^\top t, a^\top \Sigma_X a + \sigma_0^2 \exp(c^\top t)). \quad (74)$$

$c = 0$ recovers the homoscedastic baseline; $\|c\|$ controls how strongly the variance responds to t . The motivation is the one already raised in the introduction of Melnychuk et al. (2023): an estimator built around a single conditional mean and a constant-variance noise model, such as their TARNet*-based plug-in, cannot represent this dependence at all, by construction; a density estimator can.

Skewed. A linear-Gaussian mean with a standard skew-normal noise term, in Azzalini’s stochastic representation: $\delta = \lambda/\sqrt{1 + \lambda^2}$, $\varepsilon = \delta |U_0| + \sqrt{1 - \delta^2} U_1$, $U_0, U_1 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$,

$$Y \mid T = t, X = x = \xi + a^\top x + b^\top t + \omega \varepsilon. \quad (75)$$

Unlike the heavy-tailed case, this model has an *exact*, elementary closed form for $p_t(y)$, from a property specific to this representation of the skew-normal: ε is a half-normal term plus an *independent* Gaussian term, and summing an independent Gaussian ($a^\top (X - \mu_X)$, after integrating X to its marginal) with that Gaussian term simply merges into it, leaving the half-normal part untouched, so $p_t(y)$ is again a skew-normal density, with an adjusted scale and shape:

$$Y \mid \text{do}(T = t) \sim \text{SkewNormal}(\text{loc} = \xi + a^\top \mu_X + b^\top t, \omega' = \sqrt{\omega^2 + \text{var}_X}, \lambda'), \quad (76)$$

with $\text{var}_X = a^\top \Sigma_X a$, $\delta = \lambda/\sqrt{1 + \lambda^2}$, $\delta' = \delta \omega / \omega'$, and $\lambda' = \delta' / \sqrt{1 - \delta'^2}$. $\lambda = 0$ recovers the symmetric, Gaussian case; $|\lambda|$ large gives strong asymmetry. Unlike the heteroscedastic and multimodal models, whose parameters move the mean or the variance of $Y \mid T, X$ with t , λ leaves both fixed and shifts only the third moment, the same isolation the heavy-tailed model performs one moment up, for kurtosis rather than skewness — so sweeping λ tests whether a discriminant can pick up an asymmetry that a location-scale comparison would miss entirely.

A dimension-dependent pitfall. $\text{var}_X = a^\top \Sigma_X a$ appears in every closed form above, added to the model’s own “characteristic” scale (s_1, s_2 for the mixture; σ for the heavy tail; σ_0 for the heteroscedastic model; ω for the skewed one). When a and Σ_X are drawn with entries of fixed scale as $p = \dim(X)$ grows, var_X grows roughly linearly in p while the characteristic scale, fixed by design, does not: at p in the tens, var_X can already exceed the characteristic scale by one to two orders of magnitude, at which point the confounding variance dominates the sum and the very property each model was built to exhibit, bimodality, heavy tails, heteroscedasticity, skewness, becomes numerically negligible in $p_t(y)$, not because the outcome mechanism changed, but because it was diluted by an unrelated, dimension-dependent term. This effect (verified numerically, not assumed) is controlled for throughout Section 7.3 by rescaling each model’s characteristic scale in proportion to $\sqrt{\text{var}_X}$ at every value of $p = \dim(X)$ visited in the dimension sweeps, so that the targeted property remains at a comparable, detectable strength across the whole sweep rather than only at the smallest dimension.

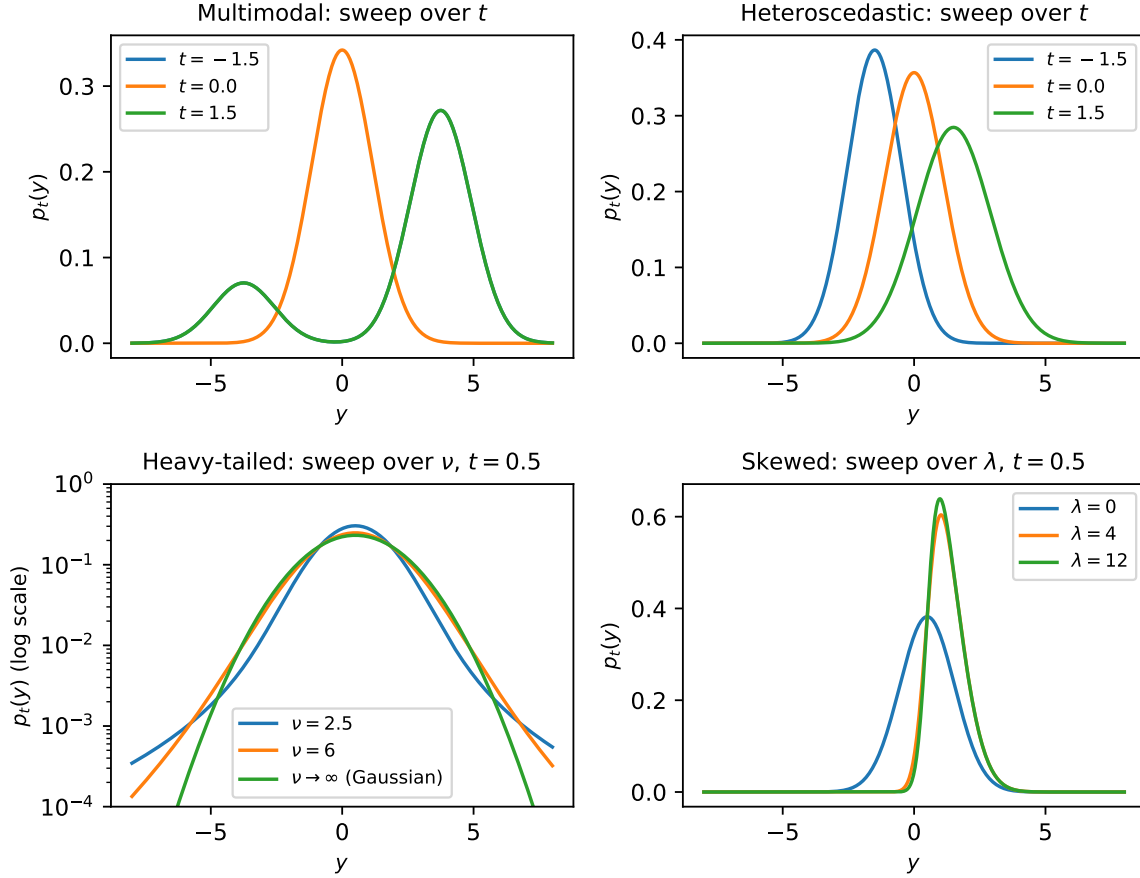


Figure 3: The four continuous, multidimensional-treatment models, illustrated at $p = \dim(X) = 1$. Multimodal and heteroscedastic panels sweep t , the property each is built to exhibit varying directly with the treatment level; heavy-tailed and skewed panels instead sweep the model's own shape parameter (ν , λ) at one fixed t , since that is what controls the departure from Gaussianity in those two cases. The heavy-tailed panel is on a log scale to make the tail behaviour, rather than the mode, visible. Parameters are illustrative only, chosen to make each property visually clear, and are not the values used in Section 7.3.

F.4 Real and semi-synthetic datasets

Four further datasets appear in Section 1 and in the companion, more applied chapter’s broader comparison against the baselines of Appendix J, but not in this paper’s own simulation study; they are documented here for completeness, since none of them is otherwise defined in the main text. All four have a binary treatment, real or semi-synthetic covariates, and simulated (hence known) potential outcomes, which is what makes them usable for interventional density estimation benchmarking despite being real-data-based. IHDP (Hill, 2011) pairs 25 real covariates from a randomized early-intervention study for low-birth-weight infants ($n \approx 747$) with two simulated response surfaces; its role in the companion chapter’s comparison is the opposite of AXY’s covariate-dimension sweep above, a small, fixed, realistic n and $\dim(X)$, where every method has enough data to be estimated reasonably well and the comparison is not dominated by a dimensionality effect. ACIC 2016 (Dorie et al., 2019) and ACIC 2018 (Shimoni et al., 2018) are collections of, respectively, 77 and 24 semi-synthetic datasets built from real covariates (linked birth and infant death records, for 2018) under a range of simulated outcome-generating mechanisms, designed as a data-analysis competition benchmark rather than a single fixed design. HC-MNIST (Jesson et al., 2021) replaces tabular covariates by 28×28 MNIST images (LeCun et al., 1998), augmented with one further hidden (unobserved) confounder, so that the covariate is high-dimensional and positivity is deliberately violated by construction.

G Cross-validation and hyperparameter grids

Every method has hyperparameters, the penalty λ , the Monte Carlo size S_{mc} , and, for the neural-network discriminant, the parameters governing the architecture itself, and all of them are selected once per configuration by a 5-fold cross-validation on the training portion of that configuration, then held fixed across the 50 replications of Section 7.3, exactly as Melnychuk et al. (2023) tune their own baselines once, on the first of their ten folds, and freeze the result for the rest.

Tuning criterion. It deliberately never touches the causal target, but not, in the end, through the contrast classifier’s own binary cross-entropy as first proposed: computing a held-out cross-entropy for $D_{\hat{\beta}_\lambda}$ would need a public scoring hook on the discriminant that the implementation does not expose. What is used instead reuses only $\hat{p}_t^{\text{ifc}}(y)$ itself, already available from (33): the held-out *factual* log-likelihood,

$$\text{log-lik}(\text{held-out fold}) = \frac{1}{n_{\text{val}}} \sum_{i \in \text{val}} \log \hat{p}_{T_i}(Y_i), \quad (77)$$

each term evaluated at the observation’s own factual (T_i, Y_i) , never at a counterfactual value. Because $\hat{p}_t(y)$ is marginal in t , it does not condition on X by construction — (77) ignores X_i entirely, an intentional weakening relative to a criterion that could condition on it (as the flow/TARNet baselines’ own nuisance tuning does); what it keeps is the property that matters here, shared with the cross-entropy originally proposed and with Melnychuk et al. (2023)’s own nuisance tuning: only observed, factual quantities are used, so tuning is a statement about how well a nuisance is estimated, never about how close the resulting causal estimate happens to land to a truth that would not be available outside a simulation. On the four data-generating processes of Appendix F.3, where the true $p_t(y)$ is available, (77) and causal accuracy are compared directly as a check that the two are not systematically at odds, but the former, not the latter, is what selects the hyperparameters actually used.

Grids. A single grid across every configuration would under-serve both ends of the dimension range this paper targets: a λ or a discriminant capacity tuned for $\dim(X) + \dim(T)$ in the tens has no reason to remain adequate in the thousands. Table 3 therefore gives three grids, indexed by $\dim(X) + \dim(T)$ of the configuration being tuned, each shifting its ranges rather than merely repeating the smaller one at a larger scale: λ ’s upper end grows with dimension, following directly from the diagnosis in Remark 1 that too small a λ is specifically dangerous under a misspecified ϕ ; the neural-network discriminant’s capacity (hidden width) grows to give it a chance of using the extra dimensions rather than being capacity-starved by a grid sized for a much smaller problem.

Method	Parameter	$\dim(X) + \dim(T)$		
		≤ 20	$(20, 200]$	> 200
Both	λ	$\{0.1, 0.3, 0.5, 1, 2\}$	$\{0.3, 0.5, 1, 2, 5\}$	$\{0.5, 1, 2, 5, 10\}$
	S_{mc}	$\{100, 200, 400\}$	$\{200, 400, 800\}$	$\{400, 800\}$
Neural-network	hidden width	$\{16, 32, 64, 32 16\}$	$\{64, 128, 64 32, 128 64\}$	$\{128, 256, 128 64, 256 128 64\}$
Logistic	feature-map degree	$\{1, 2\}$	$\{1, 2\}$	$\{1\}$ only

Table 3: Cross-validation grids for the IF-corrected estimator’s own hyperparameters, by method (“both” = logistic and neural-network alike) and by $\dim(X) + \dim(T)$ of the configuration being tuned. λ is the ridge penalty; S_{mc} the Monte Carlo size for $\hat{b}_{TX}, \hat{b}_Y$; hidden width controls k , with $a|b$ denoting one hidden layer of width a followed by one of width b . Feature-map degree 2 is dropped past $\dim(X) + \dim(T) = 200$ since the raw- X map is already the bottleneck flagged in Section 5.1 well before a quadratic expansion of it would be.

None of Table 3 is drawn from Melnychuk et al. (2023), since none of their baselines has a direct analogue among the IF-corrected estimator’s own hyperparameters, only the *protocol* (tune once, freeze, never against the causal target) is reused from them, not the grid itself. The Hessian solve inside the tuning loop is the same Cholesky factorization of Remark 1, so no grid point costs more than one ordinary fit of the estimator being tuned; even so, the neural-network cells’ full cross-product (crossing λ , S_{mc} and the hidden width) is capped at 20 combinations, a uniform random subsample taken when the full grid exceeds this, to keep the 5-fold search a small, bounded multiple of one fit rather than a multiple of the full grid size.

G.1 The baselines’ own grid

The INFS+, INFS, CNF, CNF+TS, TARNET*, MDN and DKME baselines of Appendix J are not tuned by this paper; the comparison against them belongs to the companion, more applied chapter noted in Section 1. Table 4 reproduces, for reference, the grid Melnychuk et al. (2023) tune them on (their Appendix G, Table 5), verbatim except for abbreviations. KDE is tuned by Kim et al. (2018)’s own criterion, a weighted sum of the outcome mean-squared error and the propensity cross-entropy ($\mathcal{L}_{\text{MSE}} + \alpha\mathcal{L}_{\pi}$); DKME by the mean-squared error of its ridge regression; every flow-based model by the nuisance conditional negative log-likelihood $\mathcal{L}_{\text{NLL}}$. Two second-stage components are set heuristically rather than tuned: the target flow’s number of knots is fixed to the nuisance flow’s own value (dataset-dependent, footnoted), and CNF+TS’s basis dimensionality is fixed at $d = 10$, since Melnychuk et al. found tuning it directly harmful (too small underfits, too large produces negative density values). CNF shares the nuisance-flow row of INFS, denoted $\hat{=}$ CNF in the table, since the two are the same model (Appendix J); INFS (the covariate-adjusted ablation) reuses INFS+’s grid entirely, since it is the same architecture with the one-step correction switched off, not a separately tuned model.

Model	Sub-model	Hyperparameter	Range / value
TARNET*	—	noise regularization σ_x^2	0, 0.01 ² , 0.05 ² , 0.1 ²
		noise regularization σ_y^2	0, 0.01 ² , 0.05 ² , 0.1 ²
		learning rate η	0.001, 0.005
		minibatch size b	32, 64
		tuning strategy	random grid search, 50 runs
		tuning criterion	$\mathcal{L}_{\text{NLL}}$
		train iterations	5000, SGD (momentum 0.9)
MDN	—	mixture components n_C	5, 10, 20
		noise regularization σ_x^2	0, 0.01 ² , 0.05 ² , 0.1 ²
		noise regularization σ_y^2	0, 0.01 ² , 0.05 ² , 0.1 ²
		learning rate η	0.001, 0.005
		minibatch size b	32, 64
		tuning strategy	random grid search, 50 runs
		tuning criterion	$\mathcal{L}_{\text{NLL}}$
KDE	—	learning rate η	0.001, 0.005, 0.1
		minibatch size b	32, 64, 128
		tuning strategy	full grid search
		tuning criterion	$\mathcal{L}_{\text{MSE}} + \alpha \mathcal{L}_{\pi}$
		train iterations	10000, Adam ($\beta=(0.9, 0.999)$)
DKME	—	kernel smoothness $\sigma_k = 2h_k^2$	$10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 10, 20$
		regularization ε	$10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 10$
		tuning strategy / criterion	full grid search / ridge MSE
CNF+TS	CNF	same as INFS nuisance flow ($\hat{=}\text{CNF}$)	
	TS	basis dimensionality d	10 (not tuned)
INFS+ / INFS	nuisance flow ($\hat{=}\text{CNF}$)	number of knots $n_{\text{knots},N}$	5, 10, 20
		noise regularization σ_x^2	0, 0.01 ² , 0.05 ² , 0.1 ²
		noise regularization σ_y^2	0, 0.01 ² , 0.05 ² , 0.1 ²
		learning rate η_N	0.001, 0.005
		minibatch size b_N	32, 64
		tuning strategy	random grid search, 50 runs
		tuning criterion	$\mathcal{L}_{\text{NLL}}$
		train iterations	5000, SGD (momentum 0.9)
	target flow	number of knots $n_{\text{knots},T}$	dataset-specific*
		learning rate η_T	0.005
		minibatch size b_T	64
		tuning strategy	not tuned
		train iterations	4000, Adam ($\beta=(0.9, 0.999)$)

Table 4: Hyperparameter grid of Melnychuk et al. (2023) for the baselines of Appendix J, reproduced from their Table 5 (their Appendix G) for reference; not used by this paper’s own simulation study (Table 3 above is). *Target-flow knot count: 5 on their own synthetic structural causal model (the design AXY, Appendix F.1, is adapted from), 10 on IHDP/HC-MNIST, equal to the nuisance flow’s own knot count on ACIC 2016/2018.

H Compute environment

All experiments in Section 7 run on a shared cluster. Table 5 lists the distinct hardware profiles available; the cluster is shared with other users, so queue wait time, not computation itself, can dominate the wall-clock time of a given submission, and is budgeted separately from the compute estimates below. Every replication of every configuration is one independent task in a single job array, so that the 50 replications of Section 7.3 run embarrassingly in parallel, each writing its own result file, merged only once every task has completed; no communication between tasks is needed, since replications share no state beyond their common, fixed hyperparameters.

CPU	GPU	Notes
20 cores / 40 threads	1× Tesla T4	
20 cores / 40 threads	1× Tesla P4	inference-oriented, weak float64 throughput
20 cores / 40 threads	none	targeted for CPU-only methods
40 cores / 40 threads	1× Quadro RTX4000	

Table 5: Hardware profiles available on the cluster used for this paper’s experiments.

GPU policy. Both discriminants share the same numerical core (Newton solve, Hessian, Cholesky factorization, three-channel score; Remark 1), which runs transparently on whichever device, CPU or CUDA — is resolved once at construction time. Whether that should be a GPU is decided per configuration from direct measurement, not assumed from the discriminant, since throughput and problem size interact non-monotonically. Table 6 gives the measurements behind that decision, each a single-configuration comparison on a Tesla P4-class GPU against 4 CPU cores.

Discriminant	Configuration	CPU / GPU wall-clock	Speed-up
Logistic	$\dim(X) = 10000$ ($k \approx 10^4$)	680.7 s / 235.8 s	2.9×
Neural-network	hidden width 10^4 ($k = 10001$)	305.5 s / 90.6 s	3.4×

Table 6: Measured wall-clock time, this paper’s implementation, CPU (4 cores) vs. GPU (Tesla P4-class, 7.4 GB). Where a card’s memory was exceeded, the binding constraint was the dense Hessian construction, a distinct bottleneck from the Monte Carlo channels, which are memory-capped independently of k by construction (Remark 1) and never triggered it.

The rule of thumb: below about a minute of single-core wall-clock time, kernel-launch and host-device transfer costs are not amortized and a GPU measures *slower* than the 4 CPU cores it would otherwise occupy; above that threshold, the GPU wins, by a margin that grows with the problem size k . This threshold, not the discriminant’s identity, decides the GPU policy for every cell of the dimension sweeps in Section 7.3: small- k configurations run on CPU, and the high-dimensional end of every sweep, exactly the regime Section 8.2 argues the contrastive route is meant for — runs on GPU whenever available.

I Abbreviations and notation

Table 7 lists every abbreviation used in this paper, with the place it is introduced. Abbreviations naming one of the baseline estimators compared against in Section 1 are not repeated here: Appendix J tabulates those, together with what each estimator does and where it comes from.

Abbrev.	Introduced	Meaning
<i>Method and statistical terms</i>		
MCD	Section 3.4	Marginal contrastive discrimination: the conditional-density estimator of Meziari et al. (2025), which replaces η by a univariate marginal times a density ratio
IF	Definition 3	Influence function. “IF-corrected” denotes the one-step estimator (33) built from it
IPW	Equation (11)	Inverse-probability weighting: reweighting by $1/e$, using no outcome model
A-IPTW	Equation (9)	Augmented inverse-probability-of-treatment weighting: the plug-in plus inverse-probability weighting of the outcome model’s residuals
GPS	Section 2.2	Generalized propensity score, written $e(t \mid x)$ throughout: the conditional density of the treatment given the covariates
MCFunct(π)	Definition 4	Marginal contrast function: the posterior probability $\mathbb{P}[Z = 1 \mid W]$ in the contrastive experiment at mixture ratio π
MDCond(π)	Section 3.4	The contrastive-dataset condition of Meziari et al. (2025): each observed vector labelled $Z = 1$ and κ recombined vectors labelled $Z = 0$, so that $\pi = 1/(1 + \kappa)$
GN	Equation (28)	Gauss–Newton, as in the split $H_\lambda = H_\lambda^{\text{GN}} + H_\lambda^{\text{curv}}$
KDE	Section 4.1	Kernel density estimate, used for the univariate marginal $\hat{f}_Y$
MINRES	Section 6.3	Minimal-residual iterative solver, used for possibly indefinite linear systems
SGD	Appendix G.1	Stochastic gradient descent
<i>Error criteria</i>		
RMSE	Section 7.3	Root mean squared error
MSE	Appendix G.1	Mean squared error, as the loss $\mathcal{L}_{\text{MSE}}$
NLL	Appendix G.1	Negative log-likelihood, as the loss $\mathcal{L}_{\text{NLL}}$
<i>Designs and datasets</i>		
AXY	Appendix F.1	Name of the binary-treatment, high-dimensional-covariate design adapted from the toy structural model of Melnychuk et al. (2023); not an acronym
IHDP	Appendix F.4	Infant Health and Development Program, the early-intervention study whose covariates Hill (2011) pairs with simulated response surfaces
ACIC	Appendix F.4	Atlantic Causal Inference Conference; ACIC 2016 (Dorie et al., 2019) and ACIC 2018 (Shimoni et al., 2018) are its semi-synthetic data-analysis competition benchmarks
HC-MNIST	Appendix F.4	The semi-synthetic image design of Jesson et al. (2021), which replaces tabular covariates by MNIST digit images and adds one hidden confounder
MNIST	Appendix F.4	The handwritten-digit image dataset of LeCun et al. (1998)
<i>Compute</i>		
CPU, GPU	Appendix H	Central and graphics processing unit; CUDA denotes the GPU compute platform the implementation targets

Table 7: Abbreviations used in this paper. The baseline estimators named in Section 1 (INFS+, INFS, CNF, CNF+TS, TARNET*, MDN, DKME, KDE) and this paper’s own two (MCD-PI, MCD-IFC) are listed separately in Table 10, where TS denotes the truncated orthogonal series of Kennedy et al. (2023).

Tables 8 and 9 do the same for notation, the first for the data, the target and the classical correction, the second for the contrastive construction and the discriminant fitted on it.

J Glossary of interventional density estimators

Table 10 collects, in one place, every discrete-treatment estimator named in Section 1: the abbreviation used for it throughout, what it does, which of the three families of Section 1 it belongs to, whether it corrects the first-order plug-in bias of Section 2.3 or not, and its source. All nine assume a treatment with finitely many levels and estimate the conditional outcome density directly (or, for DKME, a kernel embedding of it), which is precisely the high-dimensional, back-door object Sections 2–3 of this paper build a continuous, multidimensional-treatment alternative to.

The three families group the estimators by what they represent and how: the flow-based family represents the outcome law directly as one or two normalizing flows, and is the most flexible but also the most fully parametric, so every member can misspecify the flow’s functional form; the regression-plugin family fits a shared neural representation and reads off a fixed output family (Gaussian, or a Gaussian mixture) from it, cheaper but restricted to that family, and uncorrected, so it inherits the first-order bias of Section 2.3 exactly; and the kernel/embedding family is nonparametric in the outcome, at the price of the same curse-of-dimensionality in the conditioning set (T, X) that motivates Section 4’s use of marginal contrastive discrimination in the first place. MCD-PI and MCD-IFC, last in the table, are a fourth family of one: neither a flow, a fixed-family regression, nor a kernel estimator of the conditional density, but a reduction to a univariate density and a binary classifier, uncorrected in the first case and one-step-corrected, through its own factorization rather than through e , in the second.

Symbol	Introduced	Meaning
<i>Data and target</i>		
$W = (T, X, Y), F$	Section 2.1	An observation and its law; n independent copies are observed
$T \in \mathcal{T} \subseteq \mathbb{R}^d$	Section 2.1	Treatment, possibly a vector; $d = \dim(T)$
$X \in \mathcal{X} \subseteq \mathbb{R}^p$	Section 2.1	Covariates; $p = \dim(X)$
$Y \in \mathcal{Y} \subseteq \mathbb{R}$	Section 2.1	Scalar outcome; $Y(t)$ its potential value under $\text{do}(T = t)$
$f, \hat{f}$	Section 2.1	Densities under F and their estimates, e.g. $f_Y, f_{T,X}, f_{Y T=t, X=x}$
$p_t(y)$	Definition 1	The target: the interventional density of $Y(t)$ at y
$\eta(x)$	Equation (1)	$f_{Y T=t, X=x}(y)$, the conditional density read at the target's own (t, y)
$e(t x)$	Section 2.2	Generalized propensity score: the conditional density of T given X
$\mathcal{D}_{\text{obs}}^{T,X,Y}, \mathcal{D}_{\text{fit}}^{T,X,Y}, \mathcal{D}_{\text{eval}}^{T,X,Y}$	Section 4.1	The observations, and the two cross-fitting folds
<i>The correction</i>		
F_ε	Equation (6)	Contamination path $(1 - \varepsilon)F + \varepsilon \delta_w$
$\text{IF}(w; \psi, F)$	Definition 3	Influence function of the functional ψ at F , evaluated at w
Rem_2	Equation (7)	Second-order remainder of the von Mises expansion
δ_a	Equation (8)	Dirac measure at a
K_{h_T}, K_{h_Y}	Section 3.2	Kernels regularizing the two Dirac masses, with bandwidths h_T, h_Y
ω_i	Equation (11)	Inverse-probability weight $K_{h_T}(T_i - t)/\hat{e}(t X_i)$
$\mathcal{O}, \mathcal{O}_p$	—	Deterministic and stochastic order symbols
<i>The four estimators of $p_t(y)$</i>		
$\hat{p}_t^{\text{pi}}(y)$	Equation (3)	Plug-in: the back-door average of $\hat{\eta}$
$\hat{p}_t^{\text{ipw}}(y)$	Equation (11)	Inverse-probability weighted, using no outcome model
$\hat{p}_t^{\text{c}}(y)$	Equation (9)	Classical (A-IPTW) one-step correction of the plug-in
$\hat{p}_t^{\text{ifc}}(y)$	Equation (33)	This paper's IF-corrected estimator, through the MCD factorization

Table 8: Notation for the data, the target and the classical correction.

Symbol	Introduced	Meaning
<i>The contrastive construction</i>		
$R(t, x, y)$	Equation (13)	Density ratio $f_{T,X,Y}/(f_{T,X} f_Y)$; $\hat{R}$ its estimate
π, κ	Definition 4	Mixture ratio of the contrastive experiment, with κ recombined vectors adjoined per observed one, so $\pi = 1/(1 + \kappa)$
Z	Definition 4	Contrastive label: $Z = 1$ observed, $Z = 0$ recombined
$D_\pi(t, x, y)$	Definition 4	Marginal contrast function $\mathbb{P}[Z = 1 W]$; $\hat{D}_{\text{MCD}}$ its fitted version
$\mathcal{D}_{\text{ctr}}^{W,Z}, N_{\text{ctr}}$	Section 4.1	The contrastive dataset built from $\mathcal{D}_{\text{fit}}^{T,X,Y}$, and its size
$P_1(F), P_0(F), P_\pi$	Equation (20)	Positive law $F_{T,X,Y}$, negative law $F_{T,X} \otimes F_Y$, and their π -mixture
$m(t, y)$	Equation (17)	Covariate average $\mathbb{E}_X[R(t, X, y)]$, so that $p_t(y) = f_Y(y) m(t, y)$
<i>The discriminant and its influence function</i>		
$F_\beta, D_\beta, J_\beta$	Equation (21)	Logit, discriminant $D_\beta = \sigma(F_\beta)$ with σ the sigmoid, and logit Jacobian $J_\beta = \nabla_\beta F_\beta$
$\ell, \lambda, \mathcal{R}_\lambda$	Equation (22)	Logistic loss, ridge penalty, and the penalized contrastive risk
$\beta_\lambda(F), \hat{\beta}_\lambda$	Equation (22)	Minimizer of $\mathcal{R}_\lambda(\cdot; F)$ and its sample counterpart
u_β, v_β	Equation (24)	Gradients of the loss at labels 1 and 0
H_λ	Equation (28)	Hessian of $\mathcal{R}_\lambda$ at β_λ , split as $H_\lambda^{\text{GN}} + H_\lambda^{\text{curv}}$
$S(w; \beta)$	Theorem 1	Three-channel score $S_1 + S_0^{TX} + S_0^Y$, with $\text{IF}(w; \beta_\lambda, F) = -H_\lambda^{-1} S$
b_{TX}, b_Y, b_0	Equation (29)	The three averaged Jacobians appearing in the closed form of S
$a_\lambda(t, y), \xi(t, y)$	Corollary 1	$\mathbb{E}_X[RJ_\beta]$ and the solve $H_\lambda^{-1} a_\lambda(t, y)$
ϕ, k	Equation (34)	Fixed feature map and its dimension; $J_\beta = \phi$ for the logistic discriminant
P	Equation (36)	Number of network weights, for the neural-network discriminant
S_{mc}	Section 4.8	Monte Carlo size used for $\hat{b}_{TX}, \hat{b}_Y$

Table 9: Notation for the contrastive construction and the discriminant fitted on it. The Monte Carlo size is written S_{mc} , with the subscript, so that it is never confused with the score $S(w; \beta)$. One symbol is reused: κ denotes the number of recombined vectors per observation, whereas $\kappa_{2,Y}$ in Appendix B.3 is the second moment of the outcome kernel.

Abbrev.	Mechanism	Family	Corrected?	Source
INFS+	Two normalizing flows (nuisance + target), A-IPTW-corrected	flow-based	yes (A-IPTW)	Melnychuk et al. (2023)
INFS	Same two-flow architecture, no A-IPTW correction	flow-based	no	Melnychuk et al. (2023)
CNF	Single conditional normalizing flow, used as a plug-in	flow-based	no	Trippe and Turner (2018)
CNF+TS	Conditional normalizing flow (nuisance) + truncated orthogonal series (target), A-IPTW-corrected	flow-based	yes (A-IPTW)	Kennedy et al. (2023)
TARNET*	Shared neural representation, homoscedastic Gaussian output, plug-in	regression plug-in	no	Shalit et al. (2017)
MDN	Shared neural representation, Gaussian-mixture output, plug-in	regression plug-in	no	Bishop (1994)
DKME	Distributional kernel mean embedding of the conditional law, plug-in	kernel/embedding	no	Muandet et al. (2021)
KDE	Kernel conditional density estimate, A-IPTW-corrected	kernel/embedding	yes (A-IPTW)	Kim et al. (2018)
MCD-PI	Plug-in (3), $\hat{p}_t^{\text{pi}}(y) = n^{-1} \sum_i \hat{\eta}(X_i)$, with $\hat{\eta}$ the marginal- $\times$ -classifier conditional density estimate of Meziari et al. (2025)	contrastive	no	this paper
MCD-IFC	One-step correction of MCD-PI through its own contrastive factorization, removing e from the influence function	contrastive	yes (one-step / IF)	this paper

Table 10: Baseline interventional density estimators referenced in Section 1, with this paper’s own two estimators (MCD-PI, MCD-IFC) added as the last two rows for comparison. INFS denotes the covariate-adjusted ablation of INFS+ without its A-IPTW correction. CNF is not due to Melnychuk et al. (2023): it is the pre-existing baseline of Trippe and Turner (2018), which their own no-target-flow ablation of INFS+ happens to coincide with exactly; CNF+TS (Kennedy et al., 2023) builds on the same CNF nuisance flow, adding a truncated-orthogonal-series target and an A-IPTW correction. MCD, cited to Meziari et al. (2025) throughout this paper, names only the *conditional*-density estimator $\hat{\eta}$; it is not itself an estimator of $p_t(y)$ and so is not a row of this table. Using it for the interventional density is this paper’s own step: MCD-PI is the resulting uncorrected plug-in (Section 2.3, the back-door average (3) of $\hat{\eta}$), and MCD-IFC (Section 4) is this paper’s one-step correction of that same plug-in; both rows are therefore sourced to this paper, and only $\hat{\eta}$ itself is external.